HAL open science

Superconductivity fundamentals and Applications

Charbel Tannous

▶ **To cite this version:**

Charbel Tannous. Superconductivity fundamentals and Applications. Master. Milieux Diélectriques et Magnétiques, UBO Brest, France. 2016, pp.31. ⟨hal-04385447⟩

HAL Id: hal-04385447

https://hal.science/hal-04385447v1

Submitted on 10 Jan 2024

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.

Distributed under a Creative Commons CC BY 4.0 - Attribution - International License

Superconductivity fundamentals and Applications

C. Tannous

Université de Brest, Lab-STICC, CNRS-UMR 6285, F-29200 Brest, FRANCE

(Dated: January 9, 2024)

Superconductivity is characterized experimentally and described from the theoretical point of view based on London two-fluid model, phenomenological Ginzburg-Landau theory and finally microscopic many-body BCS theory. Roadmap toward room temperature superconductivity is briefly outlined with developments based on novel pairing different from BCS Cooper pairing and strong correlation theories in high-Tc superconductors.

PACS numbers: 74.20.Fg,74.20.-z
Keywords: BCS theory and its development,Theories and models of superconducting state

Contents

I. Characteristics of a superconductor 2
A. Perfect conductivity and perfect diamagnetism 2
B. Meissner effect: flux expulsion and levitation 2
C. Phase transition and Gap detection below critical T_c and induction B_c 3
D. Isotope effect 5

II. Nature of a Macroscopic quantum state 5
A. Distinctions between a perfect metal and a superconductor 5
B. Superconductivity and superfluidity of ^{4}He 5
C. Bose condensation 5

III. London two-fluid theory 6
A. Flux quantization 8
B. Duration of persistent currents 8

IV. Type I and Type II Superconductors 9

V. Ginzburg-Landau phenomenological theory 11

VI. Bardeen, Cooper and Schrieffer microscopic theory 14
A. Cooper pairing as an instability of the Fermi Sea 14
B. BCS Ground State 16
C. The BCS Gap 18

VII. Tunneling between metals and superconductors 20
A. MIM Tunneling 20
B. MIS Giaever tunneling 20
C. SIS Giaever Tunneling 20
D. SIS Josephson Tunneling 21

VIII. Applications of Superconductivity 22
A. Superconducting Quantum Interference Devices (SQUID) 23
B. Superconducting electronics 23
C. Superconductor-semiconductor hybrids 25

IX. High temperature Superconductivity 25
A. Crystal structure 27
B. Strong interactions and the Hubbard model 28

A. Superconductivity Timeline 29

References 30

I. CHARACTERISTICS OF A SUPERCONDUCTOR

A. Perfect conductivity and perfect diamagnetism

The electric resistivity of a superconductor is strictly zero $\rho_e = 0$ and is therefore considered as a perfect metal (with many differences discussed further below).

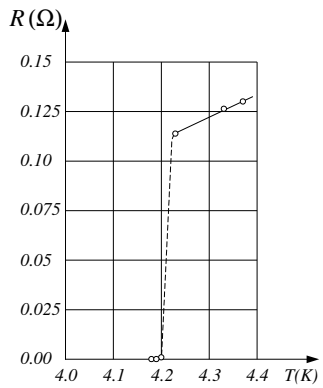

Fig.1: Kamerling-Onnes experimental work on the transition of Mercury from an ordinary metal to a superconductor at 4.2 K with a measured resistance of $10^{-5}\eta$. Adapted from Kittel (1)

After Mercury, several elements were discovered to be superconducting without or with applying pressure to them in addition to cooling, as displayed in Fig. 2.

A superconductor is a perfect diamagnet.
Note on magnetism: When a material is subjected to an external magnetic field $\boldsymbol{H}_{ext}$ corresponding, in vacuum, to a magnetic induction $\boldsymbol{B}_{ext} = \mu_0 \boldsymbol{H}_{ext}$, it acquires a magnetization $\boldsymbol{M} = \chi \boldsymbol{H}_{ext}$, so that the total magnetic induction in the material is $\boldsymbol{B} = \mu_0(\boldsymbol{H}_{ext} + \boldsymbol{M}) = \mu_0 \boldsymbol{H}_{ext}(1+\chi)$. The susceptibility χ is a scalar in isotropic materials and a tensor $\chi_{i,j}$ in anisotropic materials. For a diamagnet, the susceptibility χ is negative and for a perfect diamagnetic $\chi = -1$ implying that total magnetic induction in the material is zero.

Lenz law consequences for a perfect metal: Ohm law is written as $\boldsymbol{E} = \rho_e \boldsymbol{j}$ where $\boldsymbol{E}$ represents the electric field and $\boldsymbol{j}$ the current density. $\boldsymbol{j}$ has a finite value and since the resistivity is zero, the electric field is always screened in a perfect metal.

If an superconductor is cooled in the absence of any magnetic field and then the field is applied, when it is in the superconducting (SC) state, the following can be said. From Maxwell law $-\frac{\partial \boldsymbol{B}}{\partial t} = \boldsymbol{curl}\boldsymbol{E}$, we infer that $\boldsymbol{E}$ must remain zero in the material, $\frac{\partial \boldsymbol{B}}{\partial t} = 0$, thus the magnetic induction $\boldsymbol{B}$ should not vary. Being zero initially, it remains zero and we have perfect diamagnetism.

B. Meissner effect: flux expulsion and levitation

Suppose a YBaCuO pellet is immersed in liquid nitrogen. After reaching equilibrium at liquid nitrogen temperature (77 K), the pellet is below its critical SC temperature (90 K).

When the magnetic flux varies through the pellet, according to Lenz law, there exist currents opposing the magnetic field variation. The pellet being SC, the currents do not attenuate. The total flux through the pellet remains zero. The supercurrents create a field outside the pellet which repels the magnet, producing levitation that has many applications in cars, vehicles, trains...

After heating the YBaCuO pellet, we place a magnet on it. Then liquid nitrogen is poured. When the pellet becomes SC, the magnet floats again, which proves that during the passage of the SC transition currents originated in the pellet expel the flux of the small magnet pre-existing magnetic field.

H																He
Li	Be 0.03											B	C	N	O	F	Ne
Na	Mg											Al 1.19	Si 6.7	P 4.6–6.1	S	Cl	Ar
K	Ca	Sc	Ti 0.39	V 5.3	Cr	Mn	Fe	Co	Ni	Cu	Zn 0.9	Ga 1.09	Ge 5.4	As 0.5	Se 6.9	Br	Kr
Rb	Sr	Y 0.5–2.7	Zr 0.55	Nb 9.2	Mo 0.92	Tc 7.8	Ru 0.5	Rh 0.0003 0.49	Pd	Ag	Cd 0.55	In 3.4	Sn 3.7;5.3	Sb 3.6	Te 4.5	J	Xe
Cs 1.5	Ba 1.8;5.1	La 4.8;5.9	Hf	Ta 4.4	W 0.01	Re 1.7	Os 0.65	Ir 0.14	Pt	Au	Hg 4.15 3.95	Tl 2.39 1.45	Pb 7.2	Bi 3.9 7.2;8.5	Po	At	Rn
Fr	Ra	Ac	Ce 1.7	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu 0.1–0.7	
			Th 1.37	Pa 1.3	U 0.2	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lw	

Fig.2: . Periodic table of superconducting elements. In cyan background, the element is SC without pressure whereas a blue background means the element becomes SC after applying pressure. The critical temperatures [K] are given for each SC element. Adapted from Kittel (1).

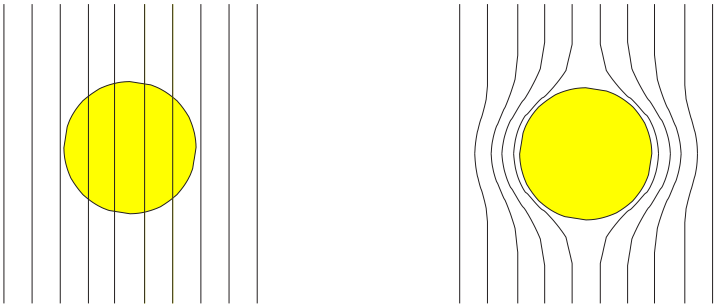

Fig.3: At left, Normal metal allowing magnetic field lines into its bulk. At right, a SC forbidding magnetic field lines. This is Meissner effect implying no coexistence of Magnetism and SC.

However, the repulsion of the magnet is weaker than during the first experiment. This SC is type II described further below. It is not as perfect as type I SC and the flux expulsion is partial.
In general, as explained further below, a magnetic field exists below the surface into a penetration depth λ_L containing a SC region.

When a metal is cooled below T_c in presence of a magnetic field $\boldsymbol{H}_{ext}$, till it becomes a perfect SC (cf Fig. 3), it expels the magnetic field with appearance of a magnetization $\boldsymbol{M}$ such that the total magnetic induction $\boldsymbol{B} = \mu_0(\boldsymbol{H}_{ext} + \boldsymbol{M}) = \mu_0 \boldsymbol{H}_{ext}(1+\chi) = 0$, meaning perfect and reversible diamagnetism.

This does not correspond to Lenz law for a perfect conductor. Indeed, $\boldsymbol{curl}\boldsymbol{E} = 0 = -\frac{\partial \boldsymbol{B}}{\partial t}$ would impose on the contrary that the magnetic induction remains at its high temperature value.

This proves the occurrence of a novel phenomenon: a phase transition from normal to SC state.

C. Phase transition and Gap detection below critical T_c and induction B_c

The Meissner effect implies the existence of a critical field. The magnetic energy necessary to expel a uniform magnetic induction B is $B^2/2\mu_0$ per unit volume. It is compensated by the difference in Helmholtz free energy between the Normal state and the SC state: $F_N(T) - F_S(T)$. This free energy difference being finite, the SC can expel, at most an induction B_c given by: $B_c^2(T)/2\mu_0 = F_N(T) - F_S(T)$.

This implies the existence of a maximum magnetic induction $B_c(T)$ above which the SC returns to normal.

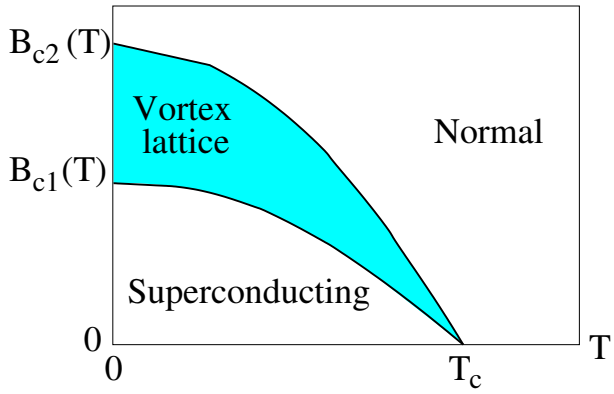

Fig.4: Phase diagram displaying the normal and SC phases in the B, T plane with $B_{c1}(T) = B_{c1}(0)[1 - (T/T_c)^2]$ and $B_{c2}(T) = B_{c2}(0)[1 - (T/T_c)^2]$ approximately observed and bounding the intermediate region containing vortex lattices in the Type II SC case.

$B_c(T)$ varies with temperature following approximately the empirical law (cf. Fig. 4): $B_c(T) = B_c(0)[1 - (T/T_c)^2]$ In particular $B_c(T)$ vanishes at the critical temperature T_c. The latent heat at the transition is expressed as: $L = T(S_N - S_S)$ where $S_N = -dF_N/dT$ and $S_S = -dF_S/dT$ represent the entropies of the normal and SC phases respectively. Using $B_c^2(T)/2\mu_0 = F_N(T) - F_S(T)$, we obtain: $L = -(TB_c/\mu_0)dB_c/dT$
It vanishes at T_c since B_c vanishes at T_c and we have a second order phase transition. At $T < T_c$ the latent heat is non-zero and we have a first order transition.

At zero field, the specific heat displays, at T_c, a jump given by the relation (1; 2):
$\Delta C_v = C_N - C_S = T\frac{d}{dT}(S_N - S_S)_{T=T_c} = -(T/\mu_0)(dB_c/dT)^2{}_{T=T_c}$. Its variation as a function of temperature is displayed in Fig. 5.

In the normal phase of a metal, the electrons are considered as a "Fermi liquid" and the specific heat is linear in T. At the transition to the SC state, a jump ΔC_v occurs in the specific heat. At very low temperatures, the specific heat is exponential $C_v(T) \approx \exp(-E_g/k_BT)$ like any system with a gap E_g in their excitation spectrum between the ground and first excited states.

In comparison with a semiconductor, this gap bears two intriguing features: It is much smaller with respect a semiconducting gap (fraction of a meV whereas for a semiconductor it is a fraction of an eV) and it is not originating from a band theory approach, i.e. it is not a single-body but a many-body effect (3).

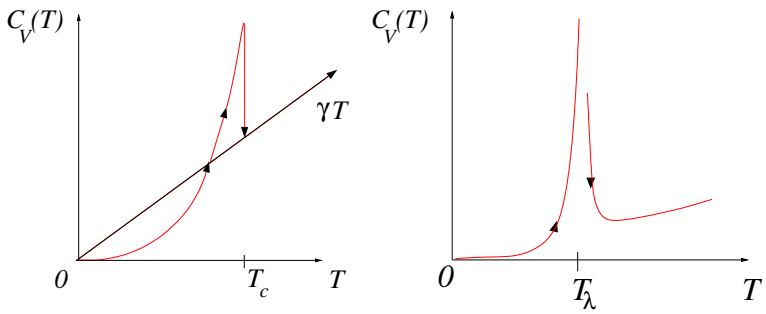

Fig.5: (Left) Illustration of the finite jump of the specific heat at a second-order phase transition temperature T_c. In a normal metal $C_v(T) = \gamma T$ like a Fermion gas where γ is Grüneisen constant whereas a SC specific heat is $C_v(T) \approx \exp(-E_g/k_BT)$ like any system with a gap E_g in its excitation spectrum. There are other possible variations of $C_v(T)$ such as a jump from a lower to a higher value at a first-order phase transition T_c. (Right) In ^{4}Helium there is a Bose-Einstein Condensation transition from normal to superfluid with $C_v(T) \sim \ln(1/|T - T_\lambda|)$ around a temperature T_λ exhibiting a lambda-like temperature dependence as if the jump of $C_v(T)$ encountered in the SC case became infinite.

D. Isotope effect

The isotope effect is a test to know whether SC is due to some interaction with the lattice. Experimentally ions in the crystal are substituted by their isotope. It is found that T_c is sensitive to isotope mass M such that $M^\alpha T_c$ is almost constant (cf Table 1).

Material	α	Material	α
Zn	0.45 ± 0.05	Ru	0.00 ± 0.05
Cd	0.32 ± 0.07	Os	0.15 ± 0.05
Sn	0.47 ± 0.02	Mo	0.33
Hg	0.50 ± 0.03	Nb_3Sn	0.08 ± 0.02
Pb	0.49 ± 0.02	Zr	0.00 ± 0.05

Table 1: Isotope effect in SCs. Experimental values of α in $M^\alpha T_c$ almost constant, with M the isotopic mass. The value of α varies considerably nevertheless, α=0.5 is the most conventional value. Adapted from Kittel (1).

II. NATURE OF A MACROSCOPIC QUANTUM STATE

A. Distinctions between a perfect metal and a superconductor

Ignoring the distinctions between a perfect metal and a superconductor would lead to pitfalls that triggered the tarnishing of the reputation of many researchers. Below is a partial list of differences:

- A metal is usually considered as an excellent conductor of electrical current and heat. In contrast, a superconductor (SC) is an an excellent conductor of electricity and a poor conductor of heat since heat destroys SC.
- Sequences of cooling a metal and a superconductor while switching on and off a magnetic field are entirely different from one another (cf. Fig. 6).
- A metal contains free single electrons (Fermion gas) whereas a superconductor contains free pairs (Cooper pairs) of electrons (Boson gas).
- Tunneling of single electrons versus electron pairs lead to tarnishing the reputation of several distinguished scientists as discussed further below.

B. Superconductivity and superfluidity of ^{4}He

It is worthy to link SC and ^{4}He superfluidity (5). Helium 4 is cooled down to about 2.1 K becomes "superfluid", i.e. its viscosity cancels out rigorously. Thus it can flow without friction (no energy dissipation), even in a very narrow capillary, similarly to a SC current circulating without Joule effect (5).

C. Bose condensation

^{4}He has nuclear spin $I = 1$ hence it is a boson (5). For an assembly of bosons, the wavefunction is totally symmetric (invariant under any particle permutation). In contrast to the fermion antisymmetric case ruled by Pauli exclusion principle, many bosons can occupy the same quantum state. At low temperature the boson system ground state is such that a large number (up to Avogadro number N_A) of particles occupy the lowest energy eigenstate yielding Bose-Einstein Condensation (5) (BEC). The BEC wavefunction represents more than the presence probability of a particle at some given point. It acquires a macroscopic meaning since we have N_A particles concerned.

Let $\psi(\boldsymbol{r})$ be the wavefunction associated with a point quantum particle and $|\psi(\boldsymbol{r})|^2$ the probability of the particle being present at point $\boldsymbol{r}$.

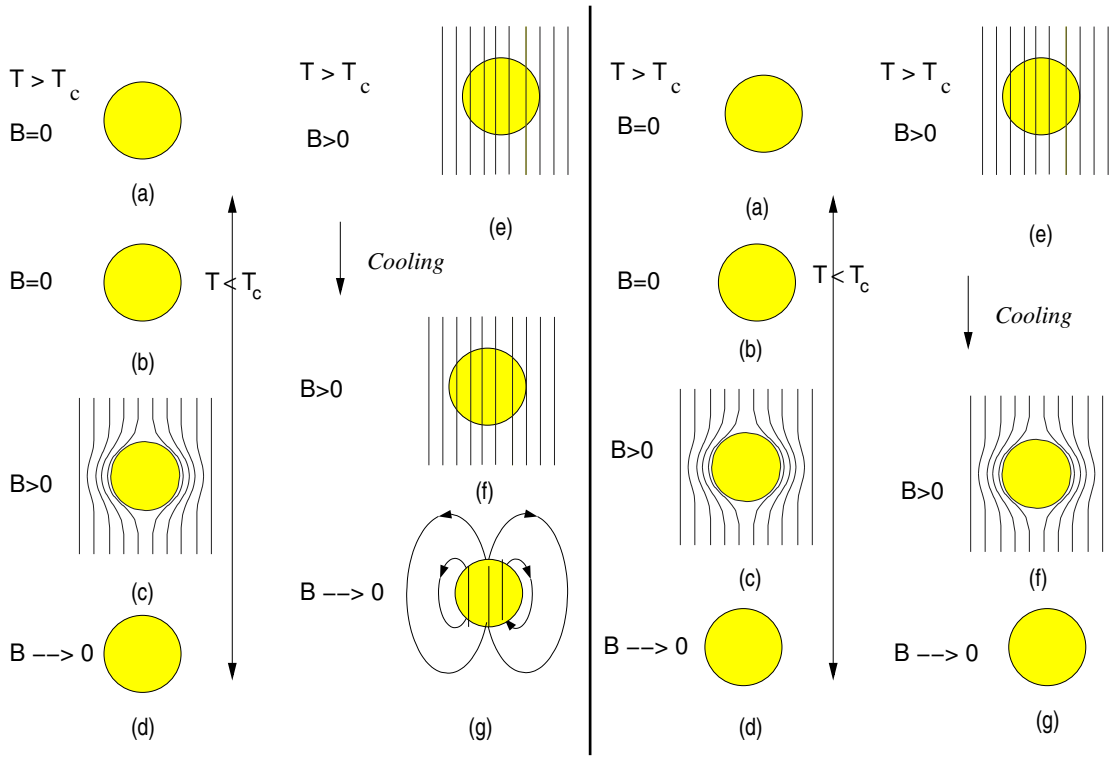

Fig.6: Magnetic behavior of a perfect metal and a superconductor: (Left panel: a-g) In a perfect metal, the final state (d) or (g) depends on whether the sample is first cooled to below T_c before applying a magnetic induction field B, or alternatively, cooled in the presence of the field. (a-b) The sample loses its resistance when cooled in zero field. (c) Application of B to sample with zero resistance. (d) Induction field B switched off. (e-f) Sample loses its resistance in the induction field. (g) Induction field B switched off. (Right panel: a-g) In a superconductor, the final states (d) and (g) are identical, regardless of whether B is switched on before or after cooling the sample: (a-b) sample loses its resistance upon cooling in the absence of an induction field. (c) Application of the field B to the superconducting sample. (d) B switched off. (e-f) Sample becomes superconducting in the applied induction field B. (g) Induction field B switched off. Adapted from Ibach *et al.* (4)

If we have a large number N_A of particles in the same quantum state $\psi(\boldsymbol{r})$ and we measure the average number of particles contained in a small but macroscopic volume $dxdydz$, it will be proportional to $|\psi(\boldsymbol{r})|^2 dxdydz$. For an assembly of condensed bosons in the same state, $|\psi(\boldsymbol{r})|^2 \propto n_s(\boldsymbol{r})$ where $n_s(\boldsymbol{r})$ is the macroscopic density of particles at point $\boldsymbol{r}$.

To explain the SC of electrons, we suppose that by associating in pairs to form spin 1 entities, they transform into bosons able to condense into the same state.

Fröhlich and Cooper (1; 6) put forward the original idea that interaction of electrons with the lattice (i.e. electron-phonon coupling) can let them overcome Coulomb repulsion to form pairs and become bosons (cf. Fig. 7).

III. LONDON TWO-FLUID THEORY

A superconductor can be simply viewed as populated by normal and superconducting electrons much like a semiconductor contains electrons and holes.
Supposing the density of normal electrons is n_N, the density of SC electrons is n_s and using Drude model we can extend the theory of normal metals to superconductors.
In the normal case, there is a damping term in the Drude model whereas it is absent in the SC case.

These assumptions can be translated straightforwardly in a quantum language. For a system of bosons all

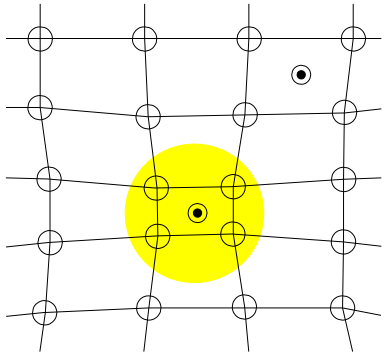

Fig.7: Distortion of the lattice illustrating electron-phonon coupling with an electron at the center of a yellow region containing a locally deformed ionic region shielding it from Coulomb repulsion with another neighboring electron through the retardation effect.

condensed in the ground state, the wavefunction becomes a macroscopic quantity: the density of particles. For a particle system charged (here Cooper pairs), by multiplying by the charge q we can associate the charge density $\rho(\boldsymbol{r})$ to the square of the amplitude of a function wave: $|\psi(\boldsymbol{r})|^2 = \rho(\boldsymbol{r})$.

The wavefunction of this "condensed" pair state can then be written: $\psi(\boldsymbol{r}) = \sqrt{\rho(\boldsymbol{r})}\exp i\theta(\boldsymbol{r})$ with amplitude field $\sqrt{\rho(\boldsymbol{r})}$ and a phase field $\theta(\boldsymbol{r})$. The transition from charge density to current density $\boldsymbol{j}$ exploits the continuity equation: $div\,\boldsymbol{j} = -\partial\rho/\partial t$. Writing the density as $\rho(\boldsymbol{r}) = \psi(\boldsymbol{r})\psi^*(\boldsymbol{r})$, we have: $\partial\rho/\partial t = \psi^*\partial\psi/\partial t + \psi\partial\psi^*/\partial t$
In the presence of a magnetic field $\boldsymbol{H} = \boldsymbol{curl}\boldsymbol{A}$, deriving from a potential vector $\boldsymbol{A}$, the Hamiltonian of the system is written, using Peierls substitution (1), as: $\hat{H} = \frac{1}{2m}(\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A})^2$.

The wavefunction ψ obeys Schrödinger equation: $i\hbar\partial\psi/\partial t = \hat{H}\psi$, thus:

$$\partial\rho/\partial t = -\frac{i}{\hbar}\psi^*[\frac{1}{2m}(\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A})\cdot(\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A}) + q\phi]\psi + \frac{i}{\hbar}\psi[\frac{1}{2m}(-\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A})\cdot(-\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A}) + q\phi]\psi^* \tag{1}$$

After expanding and rearranging we obtain:

$$\partial\rho/\partial t = -\boldsymbol{\nabla}\cdot[\frac{1}{2m}\psi^*(\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A})\psi + \frac{1}{2m}\psi(-\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A})\psi^* \tag{2}$$

Noting that $div\,\boldsymbol{j}$ is formally written $\boldsymbol{\nabla}\cdot\boldsymbol{j}$, the continuity equation becomes: $\boldsymbol{j} = \frac{1}{2m}(\psi^*(\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A})\psi) + HC)$ where HC is hermitian conjugate.
Substituting the wavefunction $\psi(\boldsymbol{r}) = \sqrt{\rho(\boldsymbol{r})}\exp i\theta(\boldsymbol{r})$, we get: $\boldsymbol{j} = \hbar\rho(\boldsymbol{\nabla}\theta - q\boldsymbol{A}/\hbar)/m$, recovering Peierls substitution.
Note that the generalized momentum $\boldsymbol{P} = \partial\mathcal{L}/\partial\boldsymbol{v}$, being derived from a Lagrangian $\mathcal{L}$, we have: $m\boldsymbol{v} = \boldsymbol{P} - q\boldsymbol{A} = \frac{\hbar}{i}(\boldsymbol{\nabla} - q\boldsymbol{A})$ allowing to write the result in the familiar form: $\boldsymbol{j} = \frac{1}{2m}\langle[\psi^*\hat{\boldsymbol{v}}\psi + HC]\rangle = \rho\boldsymbol{v}$ where $\hat{\boldsymbol{v}}$ is a velocity operator and $\boldsymbol{v}$ is its expectation value.

We consider a simply connected SC (an ellipsoid is simply connected, whereas a torus is not). and restrict ourselves to the case the density ρ is uniform.
At equilibrium, $div\,\boldsymbol{j} = 0$ because no current can emerge out of the material. Taking the divergence, we obtain: $\Delta\theta = q\,div\boldsymbol{A}/\hbar$
Adding to the vector potential the gradient of any function χ: $\boldsymbol{A} \to \boldsymbol{A} + \nabla\chi$ does not change the field $\boldsymbol{H}$ since the curl of a gradient is zero: this means a Gauge invariance symmetry. Choosing $\boldsymbol{A}$ such that: $div\,\boldsymbol{A} = 0$ and imposing the boundary condition: $A_n = 0$ meaning the vector potential $\boldsymbol{A}$ normal component on the surface is zero, defines it in a unique way.
This particular choice is called "London Gauge". For a simply connected SC, the only solution to the above is $\theta = cst$, leading to a zero gradient of θ.
Note that this is not true for a multiply connected SC.
Thus the current density is proportional to the vector potential: $\boldsymbol{j} = -q\rho\boldsymbol{A}/m$ yielding London fundamental equation.

Note: In the London Gauge, the generalized momentum: $\boldsymbol{P} = \frac{\hbar}{i}\boldsymbol{\nabla}\theta$ is zero, and reciprocally, imposing $\boldsymbol{P} = 0$ leads to both London gauge and fundamental equation.

Taking the curl of both sides of London equation, we obtain: $\boldsymbol{curl j} = -q\rho\boldsymbol{h}/m$. Using Maxwell equation: $\boldsymbol{curl h} = \mu_0\boldsymbol{j}$, we obtain: $\boldsymbol{curl}(\boldsymbol{curl h}) = -q\rho\mu_0\boldsymbol{h}/m$.

Using Maxwell equation: $div\,\boldsymbol{h} = 0$, we obtain: $\Delta\boldsymbol{h} = \boldsymbol{h}/\lambda_L^2$ with $\lambda_L = \sqrt{\frac{m}{q\rho\mu_0}}$ where λ_L is London penetration depth.

In one dimension, $\Delta\boldsymbol{h} = \boldsymbol{h}/\lambda_L^2$ is written: $d^2h(x)/dx^2 = h/\lambda_L^2$ and its solution is $h(x) = h_0\exp(-x/\lambda_L)$. The decay of the magnetic field on the surface of a SC is exponential. The field does not penetrate in the SC only over a length of the order of λ_L. It is in this penetration region below the surface where SC currents originate canceling the flux (Meissner effect). The penetration depth of London λ_L is of the order of a few tens of nanometers (cf. Table 2).

Superconductor	λ_L (nm)	ξ (nm)
Al	16	160
Cd	110	760
Pb	37	83
Nb	39	38
Sn	34	23

Table 2: Penetration Depths and Coherence Lengths at T=0K of selected superconductors. Adapted from Kittel (1).

A. Flux quantization

Let us consider a multiply connected SC (ring, torus, etc...) and imagine the sequence of steps described below:
(a) A ring of SC material is placed at room temperature in a uniform magnetic field.
(b) The ring is cooled below T_c and becomes SC. Magnetic flux lines are expelled. Magnetic field exists below the surface into a penetration depth λ_L containing SC currents. There is a magnetic flux ϕ inside the region surrounded by the ring.
(c) When the external field is removed, the flux ϕ remains constant because $d\phi/dt$ is equal to the integral of the electric field $\boldsymbol{E}$ on a closed curve inside the ring; since $\boldsymbol{E}$ is zero inside the SC: $d\phi/dt = 0$. Inside the SC ring, well beyond the penetration length λ_L, the current is zero. Thus $\boldsymbol{j} \propto (\boldsymbol{\nabla}\theta - q\boldsymbol{A}/\hbar)/m$ yielding $h\boldsymbol{\nabla}\theta = q\boldsymbol{A}$.

Consider the integral around a closed loop Γ located inside the ring, far from the surface (i.e. at a distance from the surface large with respect to λ_L), then:

$$\hbar\oint_\Gamma \boldsymbol{\nabla}\theta\cdot\boldsymbol{dl} = q\oint_\Gamma A\cdot\boldsymbol{dl} = q\int_S \boldsymbol{curl A}\cdot\boldsymbol{ds} = q\int_S \boldsymbol{B}\cdot\boldsymbol{ds} \tag{3}$$

where S designates the integral over a surface limited by the loop Γ. The right hand side is the flux ϕ of the magnetic field through the loop. We obtain: $\boldsymbol{\nabla}\theta\cdot\boldsymbol{dl} = q\Phi/\hbar$ with $q = 2e$ the absolute value of the charge of a pair of electrons (explanation is given further below). Consider two points a and b on the loop: $\oint_a^b \boldsymbol{\nabla}\theta\cdot\boldsymbol{dl} = \theta_b - \theta_a$. Returning to the same point $b = a$ after having gone through the loop, implies θ_a not necessarily equal to θ_b. This originates from the fact the wavefunction at any given point is uniquely defined: $\psi_a = \sqrt{\rho}\exp(i\theta_a) = \psi_b = \sqrt{\rho}\exp(i\theta_b)$ which imposes $[\theta_a - \theta_b]_{b=a} = 2\pi n$
This relation is expressed as: $q\phi = 2\pi n\hbar$ yielding $\Phi = n\Phi_0$ with $\Phi_0 = 2\pi\hbar/2e = h/2e$ called a fluxon.
The flux is quantized into integer values of $\Phi_0 \approx 4\times10^{-7}$ gauss.cm^2 and observed for the first time by Doll and Näbauer (7) (cf Fig. 8).

B. Duration of persistent currents

Consider a persistent current (1; 8) that flows in a SC ring. The persistent current maintains a flux through the ring of some integral number of fluxons. A fluxon cannot leak out of the ring reducing the persistent current unless

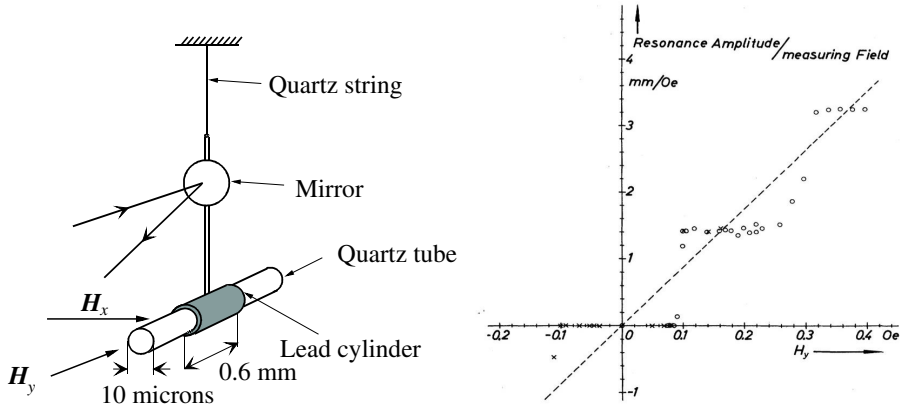

Fig.8: (Left) Doll and Näbauer setup and flux quantization steps (Right) in Lead versus magnetic field H_y with a measuring field H_x perpendicular to it. The flux trapped by the Lead sample wrapped around a quartz rod is measured from the rod oscillation period. Adapted from Doll *et al.* (7).

by a thermal fluctuation, a minimum volume of the superconducting ring, is momentarily in the normal state.
The probability per unit time (called leakage probability) that a fluxon will leak out is the product $p_L =$ (attempt frequency)(activation barrier factor). The activation barrier factor is $\exp(-\delta F/k_B T)$, where the free energy of the barrier is $\delta F =$ (minimum volume)(excess free energy density of normal state).
The minimum volume of the ring that must turn normal to allow a fluxon to escape is of the order of $R\xi^2$, where ξ is the coherence length of the superconductor and R the ring thickness. The excess free energy density of the normal state is $H_c^2/8\pi$ (1), hence the barrier free energy is $\delta F \approx R\xi^2 H_c^2/8\pi$.

Suppose the ring thickness is 1 μm, the coherence length is 1 μm, and H_c =10^3 G; then $\delta F \approx 10^{-7}$ erg. The activation barrier factor is $\exp(-\delta F/k_B T) \approx \exp(-10^{-8}) \approx 10^{(-4.34\times 10^7)}$.

The attempt frequency for the minimum volume to change its state is of the order of $\Delta/\hbar$ where Δ is the half-gap. Taking $\Delta = 1$ meV or 10^{-22} J, and $\hbar \sim 10^{-34}$ J.s, this frequency is $10^{-22}/10^{-34} = 10^{12}\text{s}^{-1}$, the leakage probability p_L becomes $p_L \sim 10^{12} 10^{(-4.34\times 10^7)} \sim 10^{(-4.34\times 10^7)}$.

The reciprocal of P is a measure of the time T_L required for a fluxon to leak out, $T_L = 1/P = 10^{(4.34\times 10^7)}$s. The age of the universe is about 13 billion years. A year is $\sim \pi \times 10^7$ s then the universe age is $\sim 13.10^9 \times \pi \times 10^7 \sim 10^{18}$ s, so that a fluxon will not leak out during the age of the universe!

IV. TYPE I AND TYPE II SUPERCONDUCTORS

A SC is characterized by two characteristic lengths (1; 6):

- The magnetic field penetration depth that varies from around 50 nm for simple metals (Al, Sn, In, Pb...) to 200 nm for transition metals or intermetallic compounds (Nb_3Sn, V_3Ga, etc...). Generally, it is 100 nm for the cuprates.
- The coherence length ξ which is the "spatial extension" of the SC pairs, is the average distance between an electron pair. Coherence length order of magnitude can be estimated from the pair bond energy Δ with Heisenberg uncertainty principle.

The characteristic wavevectors of a SC pair correspond to an energy bandwidth 2Δ centered around the Fermi energy E_F such that: $E_F - \Delta < p^2/2m < E_F + \Delta$ with $p = \hbar k$. The above relation corresponds to a momentum variation: $\delta p = 2\Delta/v_F$ where $v_F = p_F/m$ is the electron speed at the Fermi level. By Heisenberg uncertainty principle: $\delta p \delta x \approx \hbar$ and the coherence length is identified as δx: $\xi \approx \delta x \approx \hbar v_F/2\Delta$, with the approximate half-gap expression $\Delta \sim k_B T_c$ yields: $\xi \approx \hbar v_F/2k_B T_c$
The coherence length is a few nanometers for Cuprates and up to 160 nm for Aluminum. The ratio $\kappa = \lambda_L/\xi$ varies

from 0.03 for Aluminum to 100 for Y-Ba-Cu-O implying to expect a different behavior depending on whether κ is smaller or greater than 1.

For a "Type I SC", the scale over which the magnetic field $\boldsymbol{h}$ and the current density $\boldsymbol{j}$ vary, is about λ_L which is much smaller than the coherence length ξ. Conditions we used to derive the London equation are no longer valid. A more general relationship than London has been suggested by Pippard (6). It brings out another "penetration depth" λ larger than London value albeit much lower than the coherence length $\xi\kappa \gg 1$ We have a "Type II SC", or "London SC" for which London equation applies.
Examples of Type I are (Al, Sn, In, Pb...) for which $\kappa < 1$.
Intermetallic compounds Nb_3Sn, Nb_3Ge, V_3Ga,... are Type II SCs used in electromagnets and High Tc compounds (cf. Fig. 19). Their coherence length ξ is small with respect to λ_L and consequently $\kappa > 1$.

Perfect diamagnetism is observed up to relatively high induction weak B_{c1} (a few hundred Gauss), but the SC state remains up to a critical induction B_{c2} which can be very high. B_{c2} is 30 Tesla for V_3Ga and 130 Tesla for Y-Ba-Cu-O. Between B_{c1} and B_{c2}, the Meissner effect is incomplete. Here we have coexistence of normal and SC regions, but in a topology different from that encountered in the intermediate state of type I SCs. This state is called "mixed state" (cf. Fig. 9).

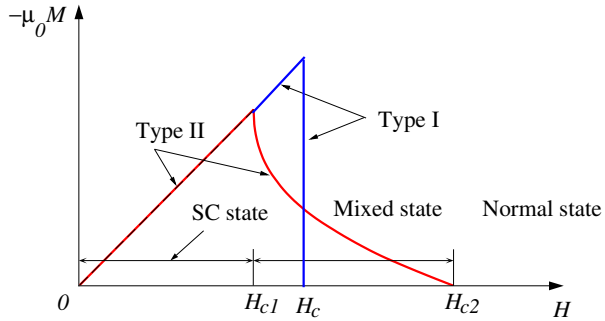

Fig.9: Variation of the magnetization $-\mu_0 M$ as a function of applied field H for Type I and Type II SCs. SC, mixed and normal states are displayed.

Defining from Thermodynamics, a critical field from: $F_N - F_S = B_c^2/2\mu_0$, we get B_c which is in general much larger than B_{c1}.
In the mixed state, a normal region displays a point defect or "Vortex" with a normal core whose radius is on the order of ξ, the coherence length. However, the magnetic field extends well beyond the normal core over a radius close to λ_L such that $\lambda_L \gg \xi$. Inside the core, the field h is equal to B_{c1}.

Actually, the flux is quantized as a multiple of the flux quantum Φ_0. Energetically, it is more favorable to create two vortices carrying each Φ_0 rather than a single $2\Phi_0$ flux, thus a vortex carries a flux quantum. In the limit $\lambda_L \gg \xi$, it is possible to derive the spatial behavior of the magnetic field within a vortex.
Outside the normal core, the field obeys London equation $-\lambda_L^2 \Delta \boldsymbol{h} + \boldsymbol{h} = 0$
Assuming the core is almost point-like (valid when $\xi \ll \lambda$), we can extend this equation to the vortex core by writing: $-\lambda_L^2 \Delta \boldsymbol{h} + \boldsymbol{h} = \boldsymbol{z} C \delta(\boldsymbol{r})$
where $\delta(\boldsymbol{r})$ is the Dirac function and $\boldsymbol{z}$ a unit vector along the vortex axis. The constant C is determined from flux quantization. Integrating each side of the previous relation on the surface of a circle enclosing the vortex yields: $C = \Phi_0$.
The analytical solution of this equation is: $h(\boldsymbol{r}) = \Phi_0/(2\pi\lambda_L^2) K_0(r/\lambda_L)$ where K_0 is the modified Bessel function of zero order and imaginary argument.
In fact, there exists a repulsive interaction between two Vortices resulting into an ordered triangular lattice.

In a type II SC, a current flows without resistance as long as the Vortices remain still. The vortices are arranged in an ordered lattice anchored to the defects and impurities of the material. However, if the current exceeds a critical value, the vortex lattice will move with some "friction" with occurrence of an electrical resistance. In intermetallic alloys, the critical current reaches large values (10,000 Amperes for Nb_3Sn). These alloys also having a very strong critical field B_{c2}, making them useful in SC magnets.
In contrast, the cuprates have low critical currents limiting their industrial applications requiring large fields.

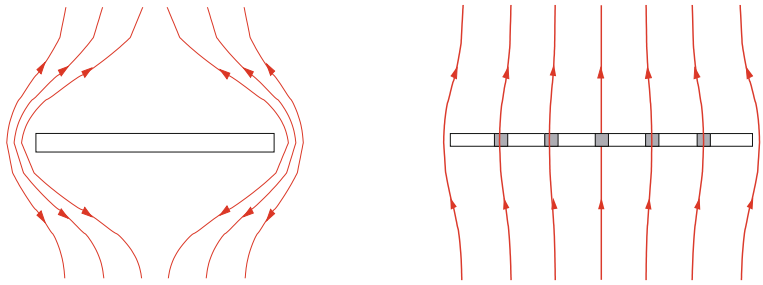

Fig.10: Magnetic field penetration patterns in Type I (Left) and Type II (Right) SC mixed state where the field penetration is present in disjoint regions.

V. GINZBURG-LANDAU PHENOMENOLOGICAL THEORY

Ginzburg and Landau had proposed an extremely useful global macroscopic approach on the basis of Landau theory of Phase Transitions.

A phase transition is characterized by a symmetry breaking (e.g. breaking of translation invariance in the liquid-solid transition) and the appearance of a new order in the system.

Landau theory of phase-transitions (2) is based on the notion of an order parameter $\boldsymbol{\eta}$ and a critical temperature T_c. When $T < T_c, \boldsymbol{\eta} \neq \mathbf{0}$, we have an ordered phase (Ferroelectric, Ferromagnetic, Superconducting...) and when $T > T_c, \boldsymbol{\eta} = \mathbf{0}$.

A general Parisi (9) prescription for a valid order parameter is:

1. Identify first the physical characteristics distinguishing the two phases of the system (normal vs superconducting).
2. Introduce an order parameter making the differences between the two phases (e. g. the density of "superconducting electrons").
3. Write a Free energy function of the order parameter; the value of the order parameter is chosen to minimize the free energy (mean field approximation).
4. Check that in the long range limit the mean field approximation gives the right answer.
5. Compute the corrections to the mean field approximation due to the finite range of the interaction.
6. If a microscopic model is available (like BCS), build from it the Landau functional with macroscopic averaging techniques and compare the results. This is exactly what Gorkov (10) did to validate BCS model and Ginzburg-Landau phenomenological model of superconductivity.

Landau hypothesis assumes that the Free energy density is a functional $F[\boldsymbol{\eta}, T]$ of the temperature dependent order parameter $\boldsymbol{\eta}$ and temperature.
Consequently, if we work around T_c we can expand the Free energy density around $\boldsymbol{\eta} \sim \mathbf{0}$ analytically in even powers of $\boldsymbol{\eta}$ such that:

$$F[\boldsymbol{\eta}, T] \sim F[0, T] + \alpha(T)[\boldsymbol{\eta}(\boldsymbol{T})]^2 + \beta(T)[\boldsymbol{\eta}(\boldsymbol{T})]^4 + ... \tag{4}$$

assuming a symmetric free energy density when $\boldsymbol{\eta}(\boldsymbol{T}) \to -\boldsymbol{\eta}(\boldsymbol{T})$.

Examples of order parameter are spontaneous magnetization $\eta = M$ for paramagnetic to ferromagnet transition, number of SC pairs $\eta \propto n_s$ in normal to SC transition etc...

At a given temperature, the order parameter is determined by the value that minimizes $F(\eta, T)$. We take $\beta > 0$ to actually obtain a minimum, rather than a maximum depending on η.

A second order transition corresponds to the facts:
- For $T > T_c$ the minimum takes place for $\eta = 0$
- For $T < T_c$ the minimum occurs for a finite value of η increasing continuously starting from 0 when moving away from T_c (cf Fig. 11).

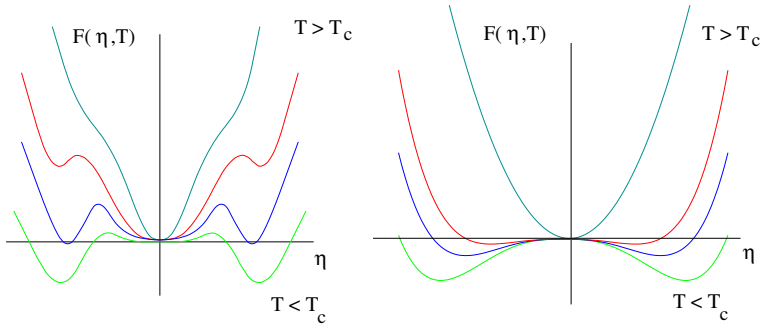

Fig.11: (Left) Variation of the free energy density $F(\eta, T)$ as a function of the order η for a first order transition (2). The minimum $\eta(T)$ is discontinuous with temperature and there is a latent heat. (Right) Variation of the free energy density for a second order phase transition (2). The minimum $\eta(T)$ varies continuously with temperature and there is no latent heat.

It follows that at T_c, the slope at the origin, given by α, changes sign, so we can write: $\alpha = \alpha_0(T - T_c)$ with $\alpha_0 > 0$.
A first order transition (cf. Fig. 11) corresponds to the following:
- At a given temperature, we have two minima, one absolute, the other relative.
- We must go beyond second order terms to account for this behavior.
- Above the critical temperature the absolute minimum corresponds to $\eta = 0$.
- At the critical temperature the two minima are degenerate.
- Below T_c, the absolute minimum occurs for a finite value of η.
- At T_c there is discontinuity in the order parameter: it is a first order transition.

In the second order case, the temperature variation of the minimum $\eta(T)$ is continuous (cf. Fig. 11).

The density of SC pairs is chosen as the order parameter $\eta = |\Psi^2| \propto n_s$ where Ψ is the complex (i.e. charge carrying) wavefunction of a Condensed Bose state. Landau and Ginzburg had an amazing intuition in describing the order parameter by the square modulus of a complex number, including the rules of gauge transformation (in the presence of a magnetic field) which are the same as for the wavefunction of a charged particle. By comparing with the experimental results, they deduced that the corresponding charge had to be $2e$ and not e, noting that nobody knew that electrons could form Cooper pairs at that particular time.

The free energy density is expanded as a power series of $|\Psi|^2$ the order parameter:

$$F(\eta, T) = F_{N0} + \alpha|\Psi|^2 + \beta|\Psi|^4/2 + \cdots \tag{5}$$

Below the critical temperature, the order parameter is: $|\Psi|^2 = -\alpha/\beta$.
The corresponding free energy density is: $F_{min} = F_{N0} - \alpha^2/2\beta$. The resulting free energy difference difference between the normal state and SC state is: $F_N - F_S = \alpha^2/2\beta = B_c^2/2\mu_0$ where B_c denotes the critical field for a type I SC and the thermodynamic field defined for a type II SC.

In a non-zero field, we must add terms depending on the magnetic field. The energy density corresponding to a magnetic field h is: $h^2/2\mu_0$. On the other hand, under certain circumstances, the density of SC pairs is not uniform within the SC (mixed state with vortices in type II SCs, intermediate states, etc.). We take also into account a spatial variation (assumed to be small) of the order parameter by a gradient of Ψ term in the free energy density.
It is imperative that this term is invariant through gauge change and this imposes a particular form on it. Indeed if we write: $\Psi = |\Psi| \exp(i\theta)$, we obtain: $\boldsymbol{\nabla}\Psi = e^{i\theta}\boldsymbol{\nabla}|\Psi| + \Psi \cdot i\boldsymbol{\nabla}\theta$
Taking $\boldsymbol{A} \to \boldsymbol{A} + \boldsymbol{\nabla}\chi$ leads to the phase change: $\theta \to \theta - \frac{q}{\hbar}\chi$ thus: $\boldsymbol{\nabla}\theta \to \boldsymbol{\nabla}\theta - \frac{q}{\hbar}\boldsymbol{\nabla}\chi$.
The most general gradient term which is invariant under gauge change gauge is then: $(\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A})$. Settling for the lowest order gradient term: $\gamma|(\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A})\Psi|^2$
There is some arbitrariness, in the free energy derivation, regarding the coefficients α, β, γ and the order parameter modulus $|\Psi|$.

We can multiply $|\Psi|$ by an arbitrary constant $|\Psi| \to c|\Psi|$ and redefine accordingly: $\alpha \to \alpha/c^2, \beta \to \beta/c^4$ and $\Gamma \to \Gamma/c^2$. In other words, among the three constants α, β and Γ we can choose only one arbitrarily.
Taking: $c = 1/2m$ yields: $\frac{1}{2m}|(\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A})\Psi|^2 = \frac{\hbar^2}{2m}(|\Psi|^2) + \frac{1}{2m}(\hbar\boldsymbol{\nabla}\theta - q\boldsymbol{A})^2|\Psi|^2$

The term $\frac{\hbar}{i}\boldsymbol{\nabla} - q\boldsymbol{A}$ is the momentum $m\boldsymbol{v}$ accounting for the Peierls substitution. The second term of the right hand side of the above equation is $\frac{1}{2m}\boldsymbol{v}^2|\Psi|^2$. It is identified with the energy of SC currents if $|\Psi|^2 = n_s/2$ is half the density of SC electrons, explaining London approach. which justifies the normalization adopted.
The free energy density takes the general form:

$$F(\eta, T) = F_{N0} + \alpha|\Psi|^2 + \beta|\Psi|^4/2 + \frac{1}{2m}|(\frac{\hbar}{i}\boldsymbol{\nabla}\theta - q\boldsymbol{A})\Psi|^2 + \boldsymbol{h}^2/2\mu_0 + \cdots \tag{6}$$

The charge q which occurs in this expression is twice the electron charge $q = 2e$ as realized by Ginzburg and Landau, when comparing to their theory. The experimental results had already predicted this. Complete justification appeared subsequently with BCS theory. In fact, in the light of the BCS theory as described below, the half-gap Δ can be chosen as an order parameter, especially since it is a complex quantity.
Integrating over the volume of the sample, the free energy becomes:

$$\mathcal{F} = \int d^3r[F_{N0} + \alpha|\Psi|^2 + \beta|\Psi|^4/2 + \frac{1}{2m}|(\frac{\hbar}{i}\boldsymbol{\nabla}\theta - q\boldsymbol{A})\Psi|^2 + \boldsymbol{h}^2/2\mu_0 + \cdots] \tag{7}$$

$\mathcal{F}[\Psi, \boldsymbol{A}]$ is a functional of Ψ and $\boldsymbol{A}$ and thermodynamic equilibrium is found by minimizing (or extremizing) $\mathcal{F}[\Psi, \boldsymbol{A}]$ with respect to both Ψ and $\boldsymbol{A}$.
The variation of $\mathcal{F}$ with respect to Ψ gives, after integration by parts (11; 12) and imposing $\delta\mathcal{F}/\delta\psi = 0$, we obtain the first Ginzburg-Landau equation (13):

$$\alpha\Psi + \beta|\Psi|^2\Psi + \frac{1}{2m}(\frac{\hbar}{i}\boldsymbol{\nabla}\theta - q\boldsymbol{A})^2\Psi = 0 \tag{8}$$

The variation of $\mathcal{F}$ with respect to $\boldsymbol{A}$, $\delta\mathcal{F}/\delta\boldsymbol{A}$ gives after integration by parts (11; 12), use of Maxwell equation: $\boldsymbol{curl}(\boldsymbol{h}) = \mu_0\boldsymbol{j}$ and imposing $\delta\mathcal{F}/\delta\boldsymbol{A} = 0$ the second Ginzburg-Landau equation (13):

$$\boldsymbol{j} = \frac{q\hbar}{2im}(\Psi^*\boldsymbol{\nabla}\Psi - \Psi\boldsymbol{\nabla}\Psi^*) - q^2\boldsymbol{A}|\Psi|^2/m \tag{9}$$

As seen at the beginning of Section. IV, two characteristic lengths were found experimentally. It becomes possible to relate them to the order parameter:
1- London penetration depth: In the case where $|\Psi|$ is uniform, thus: $|\Psi|^2 = n_s/2 = cst$.
Writing: $\boldsymbol{curl}(\boldsymbol{curlh}) = \mu_0\boldsymbol{curlj}$, we obtain: $\Delta\boldsymbol{h} = \mu_0 q^2|\Psi|^2\boldsymbol{h}/m$
Replacing $|\Psi|^2$ by its equilibrium value $|\alpha|/\beta$ the London penetration depth is expressed as: $\lambda_L = \sqrt{m\beta/q^2\mu_0\alpha}$

2- Coherence length: In zero field: $\boldsymbol{A} = 0$ and if we are sufficiently close to T_c, we may neglect the term $|\Psi|^2\Psi$ yielding: $\frac{\hbar^2}{2m}\Delta\Psi = \alpha\Psi$.
This is rewritten as: $\xi^2\Delta\Psi = \Psi$ with $\xi = \sqrt{\frac{\hbar^2}{2m|\alpha|}}$. ξ is the coherence length over which the order parameter may spatially vary.

The parameter α is function of $T - T_c$ and β can be considered constant. λ_L and ξ both vary as $(T_c - T)^{-\frac{1}{2}}$.
On the other hand, the ratio: $\kappa = \lambda_L/\xi = \sqrt{2m\beta/q^2\hbar^2\mu_0}$ called Ginzburg-Landau parameter is constant. If we take the product: $\xi\lambda_L = \hbar/q\sqrt{\beta/2\mu_0\alpha^2}$ we then obtain: $\xi\lambda_L = \Phi_0/(2\pi B_c\sqrt{2})$ where Φ_0 is the flux quantum.

Let us consider a type II SC subjected to an external magnetic field strong enough for the SC to be destroyed and let us lower the field until the nucleation of SC in the critical field B_{c2}. Under these conditions, the order parameter is weak and we can neglect the term in $|\Psi|^2\Psi$ in the first Ginzburg-Landau equation:

$$\frac{1}{2m}(\frac{\hbar}{i}\boldsymbol{\nabla}\theta - q\boldsymbol{A})^2\Psi = -\alpha\Psi \tag{10}$$

On the other hand just at the nucleation of SC, we can neglect the effect of SC currents, then the field inside the SC remains practically equal to $\boldsymbol{B_{ext}}$. In this case, the above equation is nothing else that the Schrödinger equation

for a particle of charge q in an induction field $\boldsymbol{B_{ext}}$.

Consider an infinite medium into which the particle has a constant speed v_z, performing a circular motion in the plane perpendicular to a field defining the z axis. The cyclotron frequency is $\omega_c = q\boldsymbol{B_{ext}}/m$. The energy spectrum is described by: $E_n(v_z) = m\boldsymbol{v}_z^2/2 + (n + \frac{1}{2})\hbar\omega_c$
The ground state energy is for $v_z = 0$ and $n = 0$: $E_0 = q\hbar\boldsymbol{B_{ext}}/m$.
The critical induction B_{c2} is obtained from: $-\alpha = E_0 = q\hbar B_{c2}/m$. Thus we obtain:
$B_{c2} = B_c\sqrt{m\beta/q^2\hbar^2\mu_0} = \sqrt{2}\kappa B_c$

For a type II SC, inequality $B_{c1} < B_c < B_{c2}$ is valid when $\sqrt{2\kappa} > 1$. Consequently, we obtain the Ginzburg-Landau criterion distinguishing a type I and II SC: $\kappa < 1/\sqrt{2}$: type I SC and $\kappa > 1/\sqrt{2}$: type II SC and $B_{c2} = \sqrt{2}\kappa\Phi_0/(2\pi\sqrt{2}\xi\lambda_L) = \Phi_0/(2\pi\xi^2)$, Which shows that the critical field B_{c2} corresponds to a compact neighboring of vortex cores each with an area $\approx \pi\xi^2$ carrying a flux quantum Φ_0.

VI. BARDEEN, COOPER AND SCHRIEFFER MICROSCOPIC THEORY

A. Cooper pairing as an instability of the Fermi Sea

The SC state is fundamentally different from any possible normal metallic state (ie a perfect metal at $T = 0$K) because we have at hand a many-body problem described in references such as Fetter *et al.* or Zubarev (14; 15). Thus, the transition from the normal metal state to the SC state must be a phase transition (2; 16–18).
A phase transition is accompanied by an instability of the normal state (14; 15). Cooper showed this instability as due to a small attractive interaction between two electrons above the Fermi surface.

The attraction comes from phonon exchange. The lattice deforms slowly over the electron time scale (cf. Fig. 7). It reaches its maximum distortion in a time $\tau \sim \frac{2\pi}{\omega_D} \sim 10^{-13}$s after the electron passage. During this time, the first electron has traveled $\sim v_F\tau \sim 10^8\,\frac{\text{cm}}{\text{s}} \cdot 10^{-13}\text{s} \sim$ 1000 Angströms. The positive charge of the lattice deformation can then attract another electron without feeling the Coulomb repulsion of the first electron. Due to retardation, the electron-electron Coulomb repulsion may be neglected.

The effect of the phonons is to create an attractive interaction which tends to pair time-reversed quasiparticle states. They form an antisymmetric spin singlet so that the spatial part of the wavefunction is symmetric and nodeless to take advantage of the attractive interaction.

Schrieffer (3) gave the best pictorial view of a Cooper pair by comparing the two electrons to a couple dancing over a dancing floor full of thousands of other couples. Since the coherence length $\xi \sim 1000$ Angströms is the typical distance separating the electrons in a Cooper pair it would be like the couple were dancing together in synchrony despite this very large separation leaving way to other interfering dancing couples.

Consider these two electrons, above the Fermi surface, to obey Schrödinger equation (4; 19):

$$-\frac{\hbar^2}{2m}(\nabla_1^2 + \nabla_2^2)\psi(\mathbf{r}_1\mathbf{r}_2) + V(\mathbf{r}_1\mathbf{r}_2)\psi(\mathbf{r}_1\mathbf{r}_2) = (\epsilon\ + 2E_F)\psi(\mathbf{r}_1\mathbf{r}_2) \tag{11}$$

If $V = 0$, then $\epsilon = 0$, and $\psi_{V=0} = \frac{1}{L^{3/2}}e^{i\mathbf{k}_1\cdot\mathbf{r}_1}\frac{1}{L^{3/2}}e^{i\mathbf{k}_2\cdot\mathbf{r}_2} = \frac{1}{L^3}e^{i\mathbf{k}(\mathbf{r}_1-\mathbf{r}_2)}$, where we assume that $\mathbf{k}_1 = -\mathbf{k}_2 = \mathbf{k}$.
Expanding around the $V = 0$ state, we get:

$$\psi(\mathbf{r}_1\mathbf{r}_2) = \frac{1}{L^3}\sum_{\mathbf{k}} p(\mathbf{k})e^{i\mathbf{k}\cdot(\mathbf{r}_1-\mathbf{r}_2)} \tag{12}$$

where $|p(\mathbf{k})|^2$ is the probability of finding an electron in a state $\mathbf{k}$ and the other in $-\mathbf{k}$. The sum must be bound such that

$$E_F < \frac{\hbar^2\mathbf{k}^2}{2m} < E_F + \hbar\omega_D \tag{13}$$

Considering:

$$p(\mathbf{k}) = 0 \quad \text{for} \quad \begin{cases} \mathbf{k}\ < \mathbf{k}_F \\ \mathbf{k}\ \frac{\sqrt{2m(E_F+\hbar\omega_D)}}{\hbar} \end{cases} \tag{14}$$

Schrödinger equations may be converted into a k-space equation by multiplying it by $\frac{1}{L^3}\int d^3\mathbf{r}\ e^{-i\mathbf{k}'\cdot\ \mathbf{r}}$:

$$\frac{\hbar^2 k^2}{m}p(\mathbf{k}) + \frac{1}{L^3}\sum_{\mathbf{k}'} p(\mathbf{k}')V_{\mathbf{k}\mathbf{k}'} = (\epsilon\ + 2E_F)p(\mathbf{k}) \tag{15}$$

where

$$V_{\mathbf{k}\mathbf{k}'} = \int V(\mathbf{r})e^{-i(\mathbf{k}\ -\mathbf{k}')\cdot\mathbf{r}}d^3\mathbf{r} \tag{16}$$

describes the scattering from $(\mathbf{k}, -\mathbf{k})$ to $(\mathbf{k}', -\mathbf{k}')$. It is usually approximated as a constant for all $\mathbf{k}$ and $\mathbf{k}'$ which obey scattering shell constraints and Pauli principle:

$$V_{\mathbf{k}\mathbf{k}'} = \begin{cases} -V_0 & E_F < \frac{\hbar^2\mathbf{k}^2}{2m}, \frac{\hbar^2\mathbf{k}'^2}{2m} < E_F + \hbar\omega_D \\ 0 & \text{otherwise} \end{cases} . \tag{17}$$

so

$$\left(-\frac{\hbar^2\mathbf{k}^2}{m} + \epsilon\ + 2E_F\right)p(\mathbf{k}) = -\frac{V_0}{L^3}\sum_{\mathbf{k}'} p(\mathbf{k}') \equiv\ -A \tag{18}$$

or

$$p(\mathbf{k}) = \frac{A}{\frac{\hbar^2\mathbf{k}^2}{m} - \epsilon\ - 2E_F} \qquad (\text{i.e. for } E_F < \tfrac{\hbar^2\mathbf{k}^2}{2m} < E_F + \hbar\omega_D) \tag{19}$$

Summing (4; 19) over $\mathbf{k}$

$$\frac{V_0}{L^3}\sum_{\mathbf{k}} \frac{A}{\frac{\hbar^2\mathbf{k}^2}{m} - \epsilon\ - 2E_F} = +A \tag{20}$$

or

$$1 = \frac{V_0}{L^3}\sum_{\mathbf{k}} \frac{1}{\frac{\hbar^2\mathbf{k}^2}{m} - \epsilon\ - 2E_F} \tag{21}$$

This may be converted to a density of states integral (11; 12) using the free electron dispersion relation $E(\mathbf{k}) = \frac{\hbar^2\mathbf{k}^2}{2m}$, as an approximation, in the following way.

In a continuous system the density of states $g(E)$ is defined such that the number of k states in the interval $[E, E + dE]$ is such that:

$$g_\sigma(E)dE = \sum_{\substack{n,\boldsymbol{k} \\ E \le E_{n,\sigma}(\boldsymbol{k}) \le E+dE}} \tag{22}$$

where $E_{n,\sigma}(\boldsymbol{k})$ is the dispersion relation of an electron band indexed with n, σ where $n \in \mathcal{N}$ and σ is the spin component.

The density of states (DOS) is obtained from $E_{n,\sigma}(\mathbf{k})$ with formula (1):

$$g_\sigma(E) = \frac{V}{(2\pi)^d}\oint_{E_{n,\sigma}(\mathbf{k})=E} \frac{dS_E}{|\nabla_{\boldsymbol{k}} E_{n,\sigma}(\mathbf{k})|} \tag{23}$$

with $\nabla_{\boldsymbol{k}} = \frac{\partial}{\partial \boldsymbol{k}}$ and $V = L^3$ is a 3D normalization volume as in the plane wave case: $\frac{1}{\sqrt{V}}\exp(i\boldsymbol{k}\cdot\boldsymbol{r})$. The integration element dS_E indicate the quadrature is performed over constant energy surfaces $E_{n,\sigma}(\mathbf{k}) = E$ for which the group velocity modulus is given by: $|v_g(E(n,\sigma))| = |\nabla_{\boldsymbol{k}} E_{n,\sigma}(\mathbf{k})|$.

Writing the density for a single ($n = 1$) spinless band as:

$$g(E) = 2\left(\frac{L}{2\pi}\right)^d \int_{E(\mathbf{k})=E} \frac{dS_E}{|v_g(E)|} \tag{24}$$

where the factor 2 is spin degeneracy factor $2S+1$ with $S=1/2$.
The counting of states is based on the fact allowed k values are distant by $2\pi/L$ along any direction, thus it suffices to consider a k interval corresponding to an energy interval dE and divide by $2\pi/L$ for every direction.

Thus the above sum (21) is transformed into the integral:

$$1 = V_0 \int_{E_F}^{E_F+\hbar\omega_D} g(E_F) \frac{dE}{2E-\epsilon-2E_F} = \frac{1}{2} V_0 g_0(E_F) \ln\left(\frac{\epsilon\ -2\hbar\omega_D}{\epsilon}\right) \tag{25}$$

where $g_0(E_F)$ is half the DOS at the energy E_F since we are integrating over a pair of states $(\mathbf{k}, -\mathbf{k})$ and it results from the approximation $\hbar\omega_D \ll E_F$ yielding a flat DOS around E_F.
Thus we get when $\frac{V_0}{E_F} \to\ 0$ the result:

$$\epsilon\ = \frac{2\hbar\omega_D}{1-e^{2/(V_0 g_0(E_F))}} \simeq\ -2\hbar\omega_D e^{-2/(V_0 g_0(E_F))} < 0 \tag{26}$$

B. BCS Ground State

The weak phonon-mediated attractive interaction is sufficient to trigger an instability of the Fermi sea (19), leading to the formation of a Cooper pair $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$.
The scattering (4; 19)

$$(\mathbf{k}\uparrow, -\mathbf{k}\downarrow) \to\ (\mathbf{k}'\uparrow, -\mathbf{k}'\downarrow) \tag{27}$$

yields an energy V_0 if $\mathbf{k}$ and $\mathbf{k}'$ are in the scattering shell $E_F < E_{\mathbf{k}}, E_{\mathbf{k}'} < E_F + \hbar\omega_D$. Many electrons can participate in this process and many Cooper pairs are formed, yielding an altered state or phase. Note that the energy of this state is not just $\frac{N}{2}\epsilon$ less than that of the old state, since the Fermi surface is renormalized by the birth of each Cooper pair (19).
Since pairing only occurs for electrons *above* the Fermi surface, the kinetic energy actually increases: if w_k is the probability that a pair state $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ is occupied then

$$E_{kin} = 2\sum_{\mathbf{k}} w_k \xi_k, \qquad \xi_k = \frac{\hbar^2\mathbf{k}^2}{2m} - E_F \tag{28}$$

The potential energy may be written in terms of annihilation and creation operators for the pair states labeled by $\mathbf{k}$: $|1\rangle_k$ and $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ meaning occupied whereas $|0\rangle_k$ and $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ meaning unoccupied.

This implies:

$$|\psi_k\rangle = u_k\,|0\rangle_k + v_k\,|1\rangle_k \tag{29}$$

where $v_k^2 = w_k$ and $u_k^2 = 1 - w_k$.
Then the BCS state, can be constructed according to Schrieffer (3) following nuclear physics work by considering a collection of these pairs and thus may be written as

$$|\phi_{BCS}\rangle\ \simeq\ \prod_k \{u_k\,|0\rangle_k + v_k\,|1\rangle_k\}\ . \tag{30}$$

We will assume that $u_k, v_k \in \Re$. Physically this amounts to taking the phase of the order parameter to be zero (or π), making it real.

Pauli principle leads to the fact the state $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$ can be, at most, singly occupied, thus a $(s=\frac{1}{2})$ Pauli description is possible

$$|1\rangle_k = \begin{pmatrix}1\\0\end{pmatrix}_k \qquad |0\rangle_k = \begin{pmatrix}0\\1\end{pmatrix}_k \tag{31}$$

Where σ_k^+ and σ_k^-, describe the creation and annihilation of the state $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow)$

$$\sigma_k^+ = \frac{1}{2}(\sigma_k^1 + i\sigma_k^2) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \sigma_k^- = \frac{1}{2}(\sigma_k^1 - i\sigma_k^2) = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \tag{32}$$

We have: $\sigma_k^+ \begin{pmatrix} 0 \\ 1 \end{pmatrix}_k = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and the relations $\sigma_k^+ |1\rangle_k = 0, \sigma_k^+ |0\rangle_k = |1\rangle_k$ as well as: $\sigma_k^- |1\rangle_k = |0\rangle_k, \sigma_k^+ |0\rangle_k = 0$.

The process $(\mathbf{k}\uparrow, -\mathbf{k}\downarrow) \rightarrow (\mathbf{k}'\uparrow, -\mathbf{k}'\downarrow)$, if allowed, is associated with an energy reduction V_0. In our Pauli matrix view of this process is described by operators $\sigma_{\mathbf{k}'}^+ \sigma_{\mathbf{k}}^-$, thus we obtain the Hermitian expression:

$$V = -\frac{V_0}{L^3} \sum_{\mathbf{k}\mathbf{k}'} \sigma_{\mathbf{k}'}^+ \sigma_{\mathbf{k}}^- \tag{33}$$

Thus the reduction of the potential energy is given by $\langle \phi_{BCS} | V | \phi_{BCS} \rangle$ that can be evaluated using the identities: ${}_k\langle 1|1\rangle_{k'} = \delta_{kk'}, {}_k\langle 0|0\rangle_{k'} = \delta_{kk'}$ and ${}_k\langle 0|1\rangle_{k'} = 0$

$$\langle \phi_{BCS} | V | \phi_{BCS} \rangle = -\frac{V_0}{L^3} \sum_{kk'} v_k u_{k'} u_k v_{k'} \tag{34}$$

Thus, the total energy (kinetic plus potential) of the system of Cooper pairs is

$$W_{BCS} = 2\sum_k v_k^2 \xi_k - \frac{V_0}{L^3} \sum_{kk'} v_k u_{k'} u_k v_{k'} \tag{35}$$

In order to find v_k and u_k considered as variational parameters we use $w_k = v_k^2$ and $1 - w_k = u_k^2$.
Thus we may impose this constraint by choosing

$$v_k = \cos\theta_k, \qquad u_k = \sin\theta_k \tag{36}$$

At $T = 0$, we require W_{BCS} to be a minimum.

$$\begin{aligned} W_{BCS} &= \textstyle\sum_k 2\xi_k \cos^2\theta_k - \frac{V_0}{L^3} \sum_{kk'} \cos\theta_k \sin\theta_{k'} \cos\theta_{k'} \sin\theta_k \\ &= \textstyle\sum_k 2\xi_k \cos^2\theta_k - \frac{1}{4}\frac{V_0}{L^3} \sum_{kk'} \sin 2\theta_k \sin 2\theta_{k'} \end{aligned} \tag{37}$$

$$\frac{\partial W_{BCS}}{\partial \theta_k} = 0 = -4\xi_k \cos\theta_k \sin\theta_k - \frac{V_0}{L^3} \sum_{k'} \cos 2\theta_k \sin 2\theta_{k'} \tag{38}$$

$$\xi_k \tan 2\theta_k = -\frac{1}{2}\frac{V_0}{L^3} \sum_{k'} \sin 2\theta_{k'} \tag{39}$$

Introducing the parameters $E_k = \sqrt{\xi_k^2 + \Delta^2}, \Delta = \frac{V_0}{L^3} \sum_k u_k v_k = \frac{V_0}{L^3} \sum_k \cos\theta_k \sin\theta_k$, we get:

$$\xi_k \tan 2\theta_k = -\Delta \;\Rightarrow\; 2u_k v_k = \sin 2\theta_k = \frac{\Delta}{E_k} \tag{40}$$

$$\cos 2\theta_k = \frac{-\xi_k}{E_k} = \cos^2\theta_k - \sin^2\theta_k = v_k^2 - u_k^2 = 2v_k^2 - 1 \tag{41}$$

$$w_k = v_k^2 = \frac{1}{2}\left(1 - \frac{\xi_k}{E_k}\right) = \frac{1}{2}\left(1 - \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta^2}}\right) \tag{42}$$

Making the substitutions $\left(2u_k v_k = \frac{\Delta}{E_k}, v_k^2 = \frac{1}{2}\left(1 - \frac{\xi_k}{E_k}\right)\right)$ into W_{BCS}, we obtain:

$$W_{BCS} = \sum_k \xi_k \left(1 - \frac{\xi_k}{E_k}\right) - \frac{L^3}{V_0}\Delta^2. \tag{43}$$

Comparing this to the normal state energy measured relative to E_F

$$W_n = \sum_{k<k_F} 2\xi_k \tag{44}$$

or

$$\frac{W_{BCS} - W_n}{L^3} = -\frac{1}{L^3}\sum_k \xi_k \left(1 + \frac{\xi_k}{E_k}\right) - \frac{\Delta^2}{V_0} \tag{45}$$

$$\approx -\frac{1}{2} g_0(E_F)\Delta^2 < 0. \tag{46}$$

In conclusion, SC reduces the ground state energy. This can also be interpreted as $\Delta g_0(E_F)$ electrons pairs per and volume condensed into a state Δ below E_F. The average energy gain per electron is $\frac{\Delta}{2}$.

C. The BCS Gap

The Many-body (14; 15) BCS gap 2Δ is very different from the Single-body semiconductor gap. It describes both the energy gain of the BCS state as well as its excitations. Consider the energy:

$$W_{BCS} = \sum_k 2\xi_k \frac{1}{2}\left(1 - \frac{\xi_k}{E_k}\right) - \frac{L^3\Delta^2}{V_0} = -\sum_k 2E_k v_k^4 \tag{47}$$

Recall that the probability that the Cooper state $(\mathbf{k}\uparrow, \mathbf{k}\downarrow)$ was occupied, is given by $w_k = v_k^2$. Thus the first pair breaking excitation takes $v_{k'}^2 = 1$ to $v_{k'}^2 = 0$, for a change in energy

$$\Delta E = -\sum_{k\neq k'} 2v_k^4 E_k + \sum_k 2v_k^4 E_k = 2E_{k'} = 2\sqrt{\xi_{k'}^2 + \Delta^2} \tag{48}$$

Then since $\xi_{k'} = \frac{\hbar^2 k'^2}{2m} - E_F$, the smallest such excitation is just

$$\Delta E_{min} = 2\Delta \tag{49}$$

This is the minimum energy required to break a pair, or create an excitation in the BCS ground state. It is what is measured by the specific heat $C \sim e^{-2\Delta/k_B T}$ for $T < T_c$.

In some experimental setup that adds a single or several unpaired electrons, to a SC, the additional electron cannot find readily a partner for pairing. Thus it must get into one of the previously described excited states.

For a single electron, the energy is

$$E_k = \sqrt{\xi_k^2 + \Delta^2} \tag{50}$$

For $\xi_k^2 \gg \Delta, E_k = \xi_k = \frac{\hbar^2 k'^2}{2m} - E_F$, which is just the energy of a normal metal state. Thus for energies well above the gap, the normal metal continuum is recovered for unpaired electrons.

To calculate the density of unpaired electron states, recall that the density of states was determined by counting k-states. These are unaffected by any phase transition (2; 16–18). Thus, we express the equality of number of states in d^3k as:

$$D_S(E_k)dE_k = D_N(\xi_k)d\xi_k \tag{51}$$

where D_S, D_N are respectively the superconductor, normal DOS. Note that D_N is different from the free electron DOS g_0.

In the vicinity of $\Delta \sim \xi_k, D_N(\xi_k) \approx D_N(E_F)$ since $|\Delta| \ll E_F$ (in fact: $\Delta \leq 2\omega_D$). Thus for $\xi_k \sim \Delta$

$$\frac{D_S(E_k)}{D_N(E_F)} = \frac{d\xi_x}{dE_k} = \frac{d}{dE_k}\sqrt{E_k^2 - \Delta^2} \tag{52}$$

This results into:

$$\frac{D_S(E_k)}{D_N(E_F)} = \frac{|E_k|}{\sqrt{E_k^2 - \Delta^2}}, \qquad |E_k| > \Delta, \qquad \frac{D_S(E_k)}{D_N(E_F)} = 0, \qquad |E_k| < \Delta \tag{53}$$

Δ can be evaluated as:

$$\Delta = \frac{V_0}{L^3}\sum_k \sin\theta_k \cos\theta_k = \frac{V_0}{L^3}\sum_k u_k v_k = \frac{V_0}{L^3}\sum_k \frac{\Delta}{2E_k} \tag{54}$$

$$\Delta = \frac{1}{2}\frac{V_0}{L^3}\sum_k \frac{\Delta}{\sqrt{\xi_k^2+\Delta^2}} \tag{55}$$

Convert this to sum over energy states (at $T=0$ all states with $\xi<0$ are occupied since $\xi_k = \frac{\hbar^2\mathbf{k}^2}{2m} - E_F$).

$$\Delta = \frac{V_0}{2}\Delta \int_{-\hbar\omega_D}^{\hbar\omega_D} \frac{g_0(E_F+\xi)d\xi}{\sqrt{\xi^2+\Delta^2}} \tag{56}$$

$$\frac{1}{V_0 g_0(E_F)} = \int_0^{\hbar\omega_D} \frac{d\xi}{\sqrt{\xi^2+\Delta^2}} \tag{57}$$

$$\frac{1}{V_0 g_0(E_F)} = \sinh^{-1}\left(\frac{\hbar\omega_D}{\Delta}\right) \tag{58}$$

For small Δ, we get: $\Delta \simeq 2\hbar\omega_D e^{-\frac{1}{V_0 g_0(E_F)}}$.

This expression means that T_c is controlled by Debye temperature $T_D = \hbar\omega_D/k_B$ leading to speculation about increasing T_c substantially in 1D materials by suggesting the possibility of having $T_c \propto T_F$ the Fermi temperature. This idea was extensively tested in the 1970s but did not achieve $T_c > 23$ K brickwall until the discovery of the Cuprates in 1986.

The behavior of $\Delta(T)$ can be approximated numerically by $\Delta(T)/\Delta(0) = 1.74\sqrt{(1-T/T_c)}$ for $T \sim T_c$ as displayed in Fig. 12 and compares well with many experimental results.

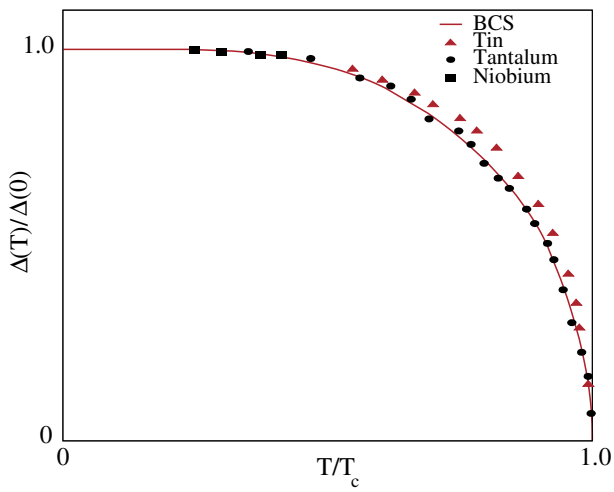

Fig.12: Variation of the BCS half-gap with temperature in comparison to experimental values pertaining to Niobium, Tantalum and Tin solids. Adapted from Serway (20)

The link between BCS and Ginzburg-Landau was made by an extraordinary *tour de force* by Gorkov (10) paving the way, for the first time, to go all the way, with Statistical Physics (2) from a microscopic theory to a macroscopic one. This helped also Nozières and Pines to relate microscopic Fermi Liquid Theory to Landau phenomenological approach (2).

Gorkov validated BCS and Ginzburg-Landau theory by linking the order parameter $|\Psi|^2$ mass m and charge q appearing in eq. 7 to Cooper pair parameters. Thus $m = 2m_e$ where m_e is the free electron mass and $q = 2e$.

VII. TUNNELING BETWEEN METALS AND SUPERCONDUCTORS

Tunneling in Superconductivity is extremely interesting since a novel phenomenon exists such as Cooper pair tunneling after bitter disputes between physicists convinced of the sole occurrence of single electron tunneling as proven by Giaever and explained by Bardeen (21). Pair and later multiparticle tunneling were developed by Schrieffer and Wilkins (22).

Let us consider a general junction made of two materials 1 and 2 separated by an insulator. When a voltage V is applied to the junction, single electron tunneling between 1 and 2 with respective Densities of States (DOS) D_1 and D_2 is described with the following currents originating from interband tunneling approach (23):

$$\begin{aligned} I_{1\to 2} &= K_0 \int_{-\infty}^{+\infty} dE\, |T_p|^2 D_1(E) f(E) D_2(E+eV)[1-f(E+eV)] \\ I_{2\to 1} &= K_0 \int_{-\infty}^{+\infty} dE\, |T_p|^2 D_1(E) D_2(E+eV) f(E+eV)[1-f(E)] \\ I = I_{1\to 2} - I_{2\to 1} &= K_0 \int_{-\infty}^{+\infty} dE\, |T_p|^2 D_1(E) D_2(E+eV)[f(E)-f(E+eV)] \end{aligned} \tag{59}$$

K_0 is a constant, T_p is the transmission coefficient across the insulating barrier that can be obtained from a semiclassical WKB approximation (2). $f(E)$ is the Fermi occupation function.

The above expressions represent a simple picture inspired from semiconductor physics (24) based on a single electron approximation whereas in the case of a superconductor, one has to consider rather a many-body approach to tunneling (4).

A. MIM Tunneling

In the case of Metal-Insulator-Metal junction, tunneling current formula (59) becomes:

$$I = K_0 |T_p|^2 D_1(E_F) D_2(E_F) \int_{-\infty}^{+\infty} dE\, [f(E) - f(E+eV)] = K_0 |T_p|^2 D_1(E_F) D_2(E_F) eV \tag{60}$$

Thus we obtain the Ohmic contact (24) case with $I \propto V$.

B. MIS Giaever tunneling

When a SC is used in a MIS junction, the question of single electrons or Cooper pairs tunneling arises. Giaever (4) treated the single electron tunneling case by the use of current formula (59) that is transformed into:

$$I = K_0 |T_p|^2 D_1(E_F) \int_{-\infty}^{+\infty} dE\, D_2(E)[f(E) - f(E+eV)] \tag{61}$$

Substituting BCS density of states formula $D_S(E)$ for $D_2(E)$, we obtain a step $V_c = \Delta/e$ in the I-V characteristic (cf. Fig. 13).

C. SIS Giaever Tunneling

In this case we consider two different superconductors with half-gaps Δ_1, Δ_2. Integrating the full current formula (59) after substituting BCS expression in the respective densities of states $D_{S1}(E), D_{S2}(E)$ (characterized by Δ_1, Δ_2) for $D_1(E), D_2(E)$ we obtain the I-V characteristic displayed in Fig. 13. It has a peak at $V_- = |\Delta_1 - \Delta_2|/e$ and a step at $V_+ = (\Delta_1 + \Delta_2)/e$.

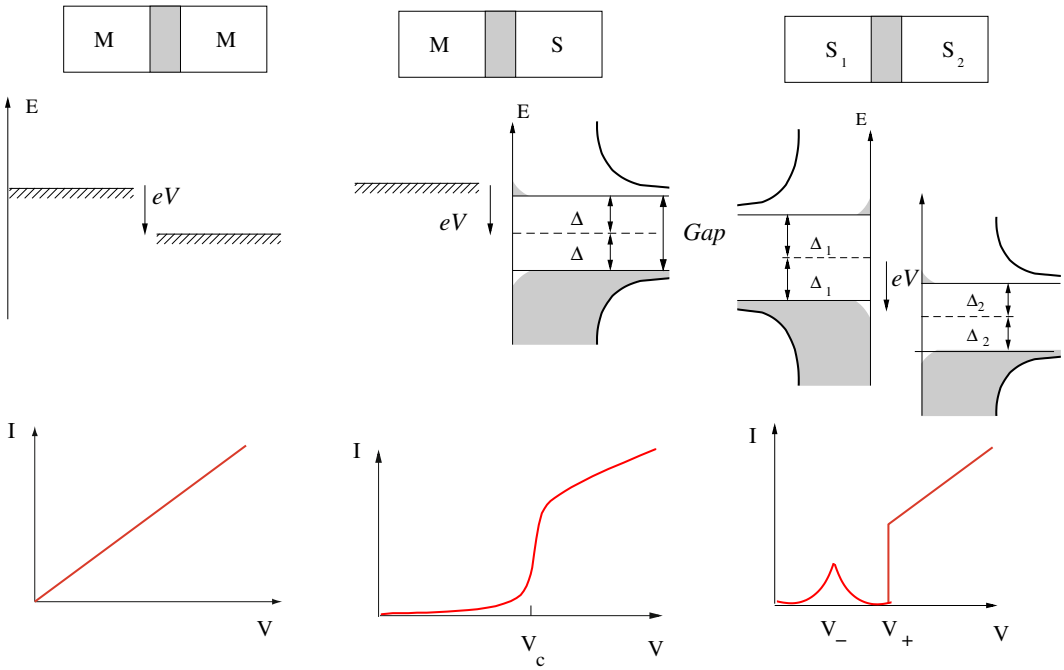

Fig.13: I-V characteristics of MIM, MIS and SIS junctions. Left: Tunneling between two metals separated by an insulator (MIM junction) Middle: Giaever tunneling between a metal and a superconductor separated by an insulator (MIS junction) with I-V characteristic displaying at T_c a step at voltage $V_c = \Delta/e$. Right: Giaever tunneling between two different superconductors separated by an insulator (SIS junction) with peak at $V_- = |\Delta_1 - \Delta_2|/e$ and a step at $V_+ = (\Delta_1 + \Delta_2)/e$. Adapted from Serway (20)

D. SIS Josephson Tunneling

Josephson considered for the first time and against many prominent physicists the case of Cooper pair tunneling instead of Giaever single electron tunneling across the insulator.
For simplicity, we assume the two identical SCs and apply a voltage V_1, V_2 at each side of the junction with ψ_1, ψ_2 the wavefunctions of the left and right SC parts of the junction. Thus we obtain the following system (25):

$$\begin{aligned} i\hbar\frac{\partial\psi_1}{\partial t} &= V_1\psi_1 + K\psi_2 \\ i\hbar\frac{\partial\psi_2}{\partial t} &= V_2\psi_2 + K\psi_1 \end{aligned} \tag{62}$$

K is the probability for an electron to cross the barrier by tunneling.
Since the voltage $V_1 - V_2 = V$ is applied to the junction, we choose the origin of the energies as $q(V_1 + V_2)/2$, transforming the system into:

$$\begin{aligned} i\hbar\frac{\partial\psi_1}{\partial t} &= \frac{q}{2}V\psi_1 + K\psi_2 \\ i\hbar\frac{\partial\psi_2}{\partial t} &= -\frac{q}{2}V\psi_2 + K\psi_1 \end{aligned} \tag{63}$$

Let us write $\psi_1 = \sqrt{\rho}_1 \exp i\theta_1; \psi_2 = \sqrt{\rho}_2 \exp i\theta_2$ and substitute into the system set.

This system can be rewritten with $\delta = (\theta_2 - \theta_1)$ as:

$$\begin{aligned} \dot{\rho_1} &= (2K/\hbar)\sqrt{\rho_1\rho_2}\sin\delta, \quad \dot{\theta_1} = (K/\hbar)\sqrt{\frac{\rho_2}{\rho_1}}\cos\delta - \frac{q}{2}V\hbar, \\ \dot{\rho_2} &= -(2K/\hbar)\sqrt{\rho_1\rho_2}\sin\delta, \quad \dot{\theta_2} = (K/\hbar)\sqrt{\frac{\rho_1}{\rho_2}}\cos\delta + \frac{q}{2}V\hbar \end{aligned} \tag{64}$$

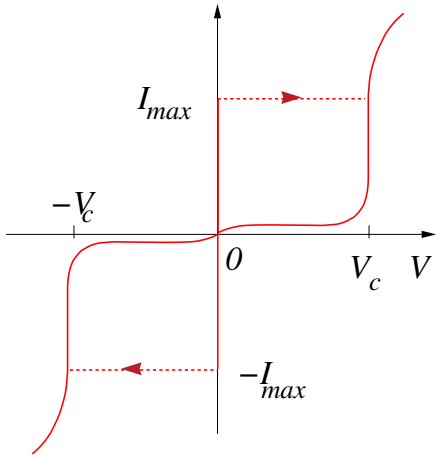

Fig.14: I-V characteristic of an SIS Josephson junction displaying switching at $V = 0$ originating from a non-zero $\pm I_{max}$ due to Cooper pair tunneling. At the critical voltage $V_c = 2\Delta/e$ where Δ is the half-gap of each superconductor, Giaever single electron tunneling is recovered. Having $\pm I_{max}$ at $V = 0$ allows information storage of a single bit: $1 \to I_{max}, 0 \to -I_{max}$. Many physicists thought that $I_{max} \neq 0$ was an experimental artifact due to junction short circuit and dismissed it. Adapted from Serway (20)

Note: In general, the variation of ρ_1 and ρ_2 are small, thus $\rho_1 \approx \rho_2 \approx \rho$, is the density of Cooper pairs.
When $V = 0$, $\dot{\delta} = (\dot{\theta}_2 - \dot{\theta}_1) = 0, \delta$ is constant and a direct current density: $J = \dot{\rho}_1 = -\dot{\rho}_2 = J_{max} \sin\delta$ can exist across the junction. $J_{max} = 2qK\sqrt{\rho_1\rho_2}/\hbar$ is the maximum current density carried by Cooper pairs and a characteristic of the junction. This effect was in fact discovered by many physicists who dismissed it thinking mistakingly that the junction was shorted.
The resulting current amplitude at $V = 0$, $I = I_{max} \sin\delta$, is phase modulated.
The I-V characteristic displays hysteresis as in Fig. 14 and the junction considered as underdamped (6) becomes a main component for storing information in switching, memory and logic circuits.

If V is a DC voltage, we have $\dot{\delta} = qV/\hbar$ thus $\delta = \delta_0 + qVt/\hbar$.
Thus an alternating current arises, oscillating very quickly, at a frequency $\omega_J = qV/\hbar = 2\pi V/\Phi_0$ called Josephson frequency. Measuring the oscillation frequency gives directly the value of the flux quantum $\Phi_0 = h/q = h/2e$.

If an alternating voltage is applied: $V = V_0 + V_1 \cos\omega_1 t$ and the phase difference across the junction $\delta = \delta_0 + \omega_0 t + (2eV_1/\hbar\omega_1)\sin(\omega_1 t)$, where $\omega_0 = qV_0/\hbar$ is Josephson frequency corresponding to V_0.

The effect of phase modulation existing in the current amplitude is to generate alternating SC currents not only at the Josephson frequency ω_0, but at frequencies $\omega_0 \pm n\omega$.
The amplitudes of these time-dependent currents are expressed in terms of ordinary Bessel functions J_n originating from sine of sine formula (11):

$$I(t) = I_{max} \sum_n (-)^n J_n(2eV_1/\hbar\omega_1) \sin(\delta_0 + \omega_0 t \pm n\omega_1 t) \tag{65}$$

If we measure the average value of the current as a function of a positive V in an I-V characteristic, these alternating currents average to zero, except for special values when $\omega_0 = n\omega_1$ i.e. when the DC voltage V_0 is at one of the values $V_n = n\hbar\omega_1/2e$.

This effect was observed for the first time by Shapiro *et al.* (26) and V_n are called Shapiro steps (1; 6) enabling to measure the flux quantum Φ_0 with high precision.

VIII. APPLICATIONS OF SUPERCONDUCTIVITY

Superconductors have potential applications in Joule-free delivery of electricity, in levitating vehicles and trains... They have applications in metrology (Volt standard definition with Josephson junction arrays (28)) and are becoming

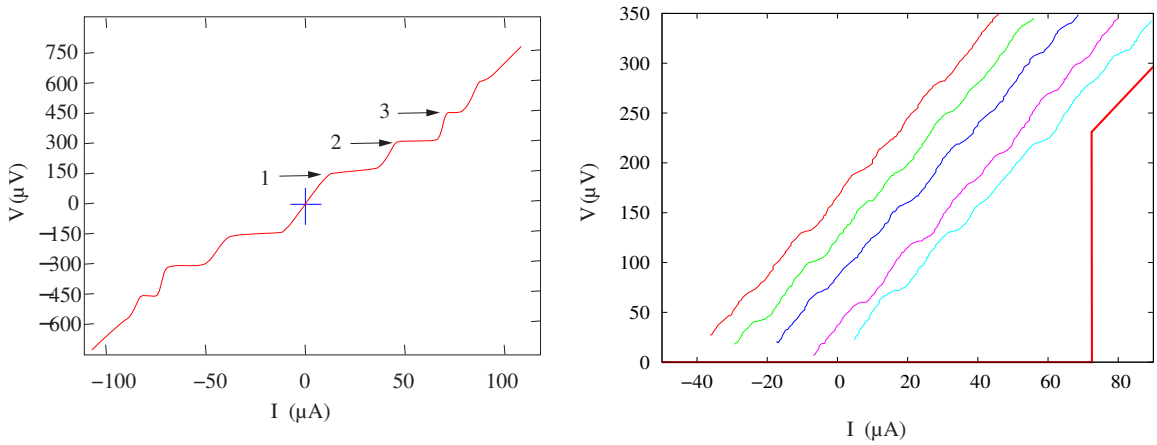

Fig.15: (Left) V-I characteristic of a Nb-Nb point contact Josephson junction with an applied 72 GHz RF voltage displaying Shapiro steps at T=4.2 K. Adapted from Ref. (26). (Right) V-I characteristics of a Nb-InAs-Nb Josephson junction, displaying Shapiro steps with an applied 15 GHz RF voltage and different powers. Wavy lines from right to left are measured with 0.165, 0.182, 0.198, 0.216, and 0.233 mW RF power, respectively at T=1.7K. When the RF power is zero, the curly structure disappears and we recover the usual flat characteristic with a step (in red at right) as in Fig. 14. Adapted from Ref. (27)

more important in many electronic applications where high frequency, high speed, low noise and low power are required.

A. Superconducting Quantum Interference Devices (SQUID)

The SQUID is made of two Josephson Junctions connected in parallel in a loop circuit. Each junction is shunted in order to eliminate switching in the I-V characteristic. SQUIDs are used as very sensitive magnetometers (29), fast samplers, shift-registers, A/D converters (cf Table 3)...

Consider two Josephson junctions a and b in parallel in an induction field $\boldsymbol{B} = \boldsymbol{curl}\boldsymbol{A}$ and let J_a and J_b be the SC current densities through the junctions and δ_a, δ_b their respective phases (cf Fig. 16).

The phase difference of the wavefunction between P and Q (cf Fig. 16), traveling through a path Γ_a which passes through a is: $\delta_{P\to Q} = \delta_a + \frac{q}{\hbar}\oint_{\Gamma_a} \boldsymbol{A}\cdot\boldsymbol{dl}$

The phase difference between P and Q, traveling a path Γ_b which passes by b is: $\delta_{P\to Q} = \delta_b + \frac{q}{\hbar}\oint_{\Gamma_b} \boldsymbol{A}\cdot\boldsymbol{dl}$

Γ being a closed loop through the system, we have: $\delta_b - \delta_a = \frac{q}{\hbar}\oint_{\Gamma} \boldsymbol{A}\cdot\boldsymbol{dl} = \frac{q}{\hbar}\Phi = 2\pi\Phi/\Phi_0$

Calling the average $\delta_0 = (\delta_a + \delta_b)/2$, we obtain: $\delta_a = \delta_0 - \pi\Phi/\Phi_0$ and $\delta_b = \delta_0 + \pi\Phi/\Phi_0$

The total current density through the device is: $J_T = J_a + J_b$ given by:

$$J_T = J_{max}[\sin(\delta_0 + \pi\Phi/\Phi_0) + \sin(\delta_0 - \pi\Phi/\Phi_0)] = 2J_{max}\sin\delta_0\cos(\pi\Phi/\Phi_0) \tag{66}$$

If we vary the external field, the Josephson current oscillates as a function of the flux with period $2\Phi_0$. This device called SQUID has steady current enabling to detect flux variations of the order of Φ_0. There are many variations of SQUIDs operating with direct or alternating current at various frequencies (29).

B. Superconducting electronics

Superconductivity is extremely interesting from the device point of view as witnessed by the value of the Josephson Junction power delay product (cf Fig. 17) being the smallest with respect to all conventional microelectronic devices.

Superconductivity electronics may be conveniently divided into five broad categories:

1. Analog detection of signals (1Hz-100GHz). At frequencies up to several hundred GHz, SIS mixers are the most sensitive detectors available.

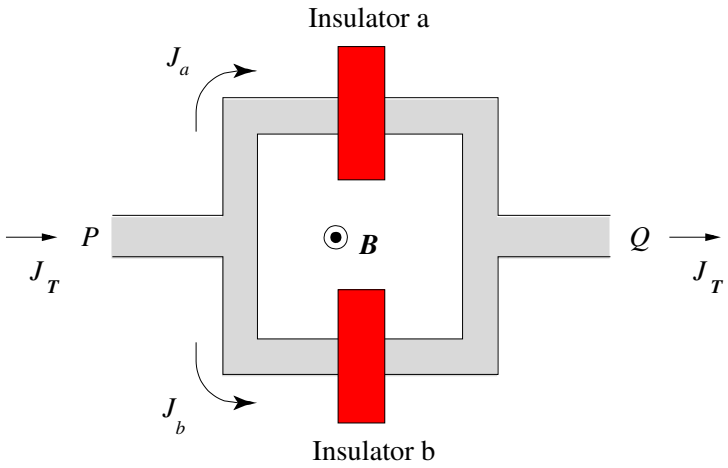

Fig.16: SQUID made of two Josephson junctions arranged in parallel in a superconducting loop and subjected to an induction field $\boldsymbol{B} = \boldsymbol{curl A}$ used in text. Adapted from Kittel (1)

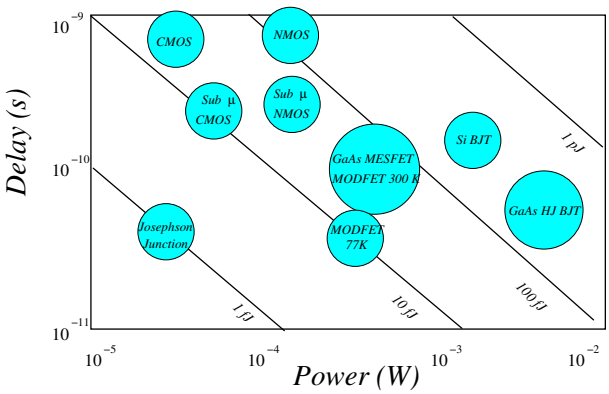

Fig.17: Power Delay product for several standard microelectronic devices compared to the Josephson Junction. HJ is heterojunction. BJT is Bipolar Junction Transistor. CMOS is Complementary Metal Oxide Semiconductor. NMOS is n-type Metal Oxide Semiconductor. FET is Field Effect Transistor. MODFET is Modulation Doped FET. MESFET is MEtal Semiconductor FET. fJ is Femto-Joule or 10^{-15} Joule, the order of magnitude of the power-delay of the Josephson Junction. pJ is pico-Joule or 10^{-12} Joule. Note that the Submicron devices (denoted as Sub μ) are more efficient than their micronic counterparts. Adapted from Ibach (24)

2. Analog processing of high frequency (GHz) signals. It is predicted that bandwidths as large as 10GHz and computational rates equivalent to 1012 arithmetic operations per second, should be possible in the near future. Very high speed samplers are based on the rapid switching of the Josephson junction. Sampling resolution times as low as 2 picoseconds have been achieved with Josephson based oscilloscopes. An even higher sampling resolution has been obtained by combining optoelectronic techniques with a SC transmission line. Pulses of 0.9 picosecond have been generated and detected with this technique. Fast A/D based on Josephson technology have been developed capable of digitizing 6 bits at 100MHz and 3 bits at 500MHz. It is projected that the next performances will be 6 bits at 300 MHz and 4 bits at 1 GHz.

3. In digital processing some demonstrations of multi GHz shift-registers have been reported and the projections are for clocking speeds of 60 GHz.

4. Sensing of very weak magnetic fields ($< 10^{-14}$ Tesla) making them useful in MEG (Magneto-Encephalo-Graphy), MCG (Magneto-Cardio-Graphy), Highly sensitive Seismic detection (Magnetometers, Gradiometers)... (cf Table 3).

5. Ordinary and Quantum Computers (Josephson computers) due to their very low power-delay product and high speed processing.

Sensing/Detection	Analog Processing	Digital Processing
DC SQUIDs	Josephson Devices	Logic
RF SQUIDs	A/D Converter	Memory
Microwave-Submillimeter	Sampler	Ordinary Computer
Josephson Mixer	Three-Terminal Devices	Quantum Computer
Josephson SLED	Superconducting FETs	
Josephson Parametric Amplifier	Matched Filters	
SC Bolometer	Transmission Line Devices	
Magnetometers, Gradiometers	Non-equilibrium Devices	
	Signal Processors	
	Convolvers	Magnetic Resonance Imaging
	Correlators	Magneto-Encephalo-Graphy (MEG)
	Fourier Transformers	Magneto-Cardio-Graphy (MCG)
	Optoelectronics	Seismic detection

Table 3: Applications of Superconductors. SLED is Square Law Envelope Detector used in Telecommunications. FET is Field Effect Transistor.

Josephson Junctions are made of two SC layers separated by an insulator. With no voltage applied a DC current exists across the junction due to the tunneling of pairs of electrons (Cooper Pairs). The I-V characteristic of the junction signifies information storage with $1 \rightarrow I_{max}, 0 \rightarrow -I_{max}$ at $V = 0$ (cf Fig. 14). Thus a Josephson junction is the basis of computer memories, logic devices and Josephson computers.

The SIS is a Josephson Junction with an applied magnetic field that destroys the DC current at zero voltage. One exploits the sharp non-linearity in the I-V characteristic (jump at voltages on the order of several mV) for making mixers and square law envelope detectors working at several 100 GHz with very low noise.

C. Superconductor-semiconductor hybrids

It would be very advantageous to use the low microwave losses and low dispersion properties of passive SC devices in combination with semiconductor transistors. Another form of hybrid would involve the combinations of active semiconductor devices (transistors) with active SC devices (Josephson junctions) on the same chip.

These applications are interesting with high temperature SCs (28). The problems to overcome are the heat produced by the semiconductor devices and the interfacing between the low voltages needed in SCs and the higher ones applied to semiconductor transistors.

The applications of the hybrid technology are listed below:

Many hybrids have already been produced, their quality is still low because of the problems mentioned above. It is expected, nevertheless that these hybrids will be very important in the next couple of years once the HTS technology is mature enough.

IX. HIGH TEMPERATURE SUPERCONDUCTIVITY

Traditional superconductors should be cooled at very low temperatures (about 4.2K) with liquid ^{4}He which is their main disadvantage.

In 1986 a family of ceramic compounds going by the name 123 because of the representative material Y(1)Ba(2)Cu(3)O(7) was discovered after the long-standing 23 K T_c brickwall encountered in intermetallic compounds.

T_c was increased later to 125 K in bulk superconducting oxides. These compounds succeeded all standard superconductivity tests: the Meissner effect, AC Josephson effect, long duration persistent currents and substantially zero DC resistivity with high transition temperatures, accompanied with high critical currents and magnetic fields.

They can operate at much higher temperatures (77K ... 138K) using liquid nitrogen which is cheap and easy to use (cf. Table 5, Table 6 and Fig. 19).

Passive HTS films	Active Josephson devices		
Computer interconnects Antennas, filters Resonators for MMW			
	Tunnel Junction		HTS conductor coupled
	LTS	HTS	LTS or/and HTS
Phase shifter Tapped delay-line Signal processor Device electrodes	Millimeter wave receiver with IF amplifier 4K logic and 27-77K semiconductor memory	Wideband JJ mixer in hybrid receiver Computer interconnects	Picosecond sampler Digital logic Digital memory Josephson OM in hybrid receiver A/D

Table 4: Hybrid Semiconductor-Superconductor devices. MMW is Microwave and Millimeter Waves. JJ is Josephson Junction. OM is Oscillator and Mixer, LTS is Low Temperature Superconductor, HTS is High Temperature Superconductor. IF is Intermediate Frequency.

Formula	Family	Highest Tc reached [K]
RE-$Ba_2Cu_3O_7$	RE-BCO or 123	92 (YBCO)
$Bi_2Sr_2Ca_{n-1}Cu_nO_{2n+4}$	BSCCO or	90 (Bi-2212)
(with Pb doping)	Bi-22 (n-1)n	122 (Bi-2223)
		90 (Bi-2234)
$Tl_2Ba_2Ca_{n-1}Cu_nO_{2n+4}$	TBCCO or	110 (Tl-2212)
	Tl-22 (n-1)n	127 (Tl-2223)
		119 (Tl-2234)
$TlBa_2Ca_{n-1}Cu_nO_{2n+3}$	Tl-12 (n-1) n	90 (Tl-1212)
(A = Sr, Ba)		122 (Tl-1223)
		122 (Tl-1234)
		110 (Tl-1245)
$HgBa_2Ca_{n-1}Cu_nO_{2n+2}$	Hg-12 (n-1)n	96 (Hg-1201)
		128 (Hg-1212)
		135 (Hg-1223)

Table 5: General families of high temperature superconductors. According to their composition their family names are e.g. (Bi-2212). RE = Rare Earths: Y; Eu; Gd... Adapted from Ibach *et al.* citeIbach.

Thin films of the Cuprates have microwave losses at 77K, much below those of copper at the same temperature. This paves the road for applications in antennas (microstrip, microwave and millimeter wave) and transmission lines for fast propagation of signals at reduced ohmic losses and near elimination of dispersion.
Frequency selective components (filters and resonators) can be made with thin films and these are of use potentially in mobile communications in the frequency range 900 MHz to 1.8 GHz.

Resonators made with the Cuprates produced radio signals 10 to 100 times sharper than anything their all-copper counterparts can create. Helical resonators can be made compact enough to be used in the repeater stations for mobile telephones.

A 4-watt, 10 MHz SC resonator fits into one hand with the production of 3D multilayer structures whereas an all-copper version would have been the size of a barrel. A SC transistor has been built at Sandia Laboratories operating at 100 GHz. Microwave amplifiers, oscillators, phase shifters and mixers have also been made. Microwave circuits are miniaturized such that they are eventually integrated on a single chip or on a system-on-chip (SOC) and used in wireless communication devices such as transmitters, modulators, receivers, car keys, smartphones...

Compound	T_c	Family
$BaPb_{0.75}Bi_{0.25}O_3$	12 K	[BPBO]
$La_{1.85}Ba_{0.15}CuO_4$	36 K	[LBCO]
$YBa_2Cu_3O_7$	90 K	[YBCO]
$Tl_2Ba_2Ca_2Cu_3O_{10}$	120 K	[TBCCO]
BiSrCaCuO	125 K	[BSSCO]
$Hg_{0.8}Tl_{0.2}Ba_2Ca_2Cu_3O_{0.33}$	138 K	[HTBCCO]

Table 6: Particular cases of Ceramic Cuprate compounds with their respective composition, T_c and abbreviated family names. Note that B could be Ba or Bi, whereas C is Cu or Ca.

A. Crystal structure

A first example is the $La_{2-x}Sr_xCuO_4$ family discovered by Berdnoz and Müller (20). The base compound La_2CuO_4 is insulating. Along the c axis, there is a periodic sequence of planes: a CuO_2 plane, two LaO planes, a CuO_2 plane and two LaO planes.
Each CuO_2 unit (cf. Fig. 18) has a Perovskite structure consisting of a tetrahedron with an Oxygen at each apex and a Copper at the center. The tetrahedron fits inside a cube containing a Lanthanum at each apex. Oxygens are in the state O^{2-} with a $2p^6$ full subshell. Cu atoms are in the Cu^{2+} state with 9 electrons (i.e. one electron is missing) in the 3d subshell.
Under the effect of an octahedral crystal field, there is lifting of degeneracy into a triply degenerate t_{2g} state, of lower energy and a doubly degenerate level a_{1g}.
Under the effect of a tetrahedral symmetry, the level t_{2g} separates into a doubly degenerate level and a single level. a_{1g} separates in two sub-levels.

There is hybridization between the highest Cu^{2+} occupied level with $3d_{x^2-y^2}$ orbitals and the Oxygen $2p$ orbitals, therefore a valence band is half-full, then according to band theory this compound should be a conductor whereas it is actually an insulator.

Band theory does not account for electron interactions. If the repulsive Coulomb interaction between electrons is strong, there can be localization of electrons and formation of a "Mott insulator" which is encountered in several cuprates and indicative of strong correlation effects (30).

Electron localization is based on the interplay between band energy and Coulomb repulsion between neighboring electrons. Band energy is close to kinetic (neglecting somehow crystal potential) and given by $E_K = \hbar^2\pi^2/(2m^*a^2)$ per particle, where a is the distance between neighboring electrons in a crystal (with lattice parameter $\sim a$) and m^* their effective mass. Coulomb repulsion between two neighboring electrons is: $U \approx e^2/(4\pi\epsilon_r\epsilon_0 a)$.

When the ratio: $U/E_K \geq r_c$ a critical value (about 50), a transition to Mott insulator results implying that large Coulomb repulsion may induce charge localization. This metal to insulator phase change is different from Mott transition that results from the interplay between electron density and screening in moderate Coulomb interaction case (4; 31).
Coulomb interaction causes folding of the Brillouin zone: Lanthanum atoms are trivalent (La^{3+}) and ionic balance is preserved. If a divalent strontium atom Sr^{2+} is substituted to a trivalent Lanthanum atom (La^{3+}), then going from Cu^{2+} to Cu^{3+} leads to creation of a hole in the upper $3d$ Copper band. This is the reason why for some substitution value x the compound $La_{2-x}Sr_xCuO_4$ becomes a conductor.
The crystal structure of SC cuprates such as $Bi_nSr_nCa_{n-1}Cu_nO_{2n+4}$ is characterized by the presence of CuO_2 planes. The $n = 1$ compound includes a CuO_2 plane surrounded by two strontium-oxygen and bismuth-oxygen planes. Its T_c =10K. The $n = 2$ compound is built from $n = 1$ by inserting a structure made of two CuO_2 planes, separated by a calcium plane, and its T_c =85 K. Replacing each CuO_2 plane with this structure, the $n = 3$ compound is obtained with T_c =110K.
CuO_2 planes are separated by other ions whose essential role is to determine the number of holes in the CuO_2 planes. We can then vary the number of holes by changing the chemical composition (by substitutions, varying oxygen atom numbers...).
In the CuO_2 plane, the electrons participating to conductivity originate from the $3d^9$ shell characterized by $d_{x^2-y^2}$ orbitals. They overlap with $2p_x$ and $2p_y$ Oxygen orbitals.

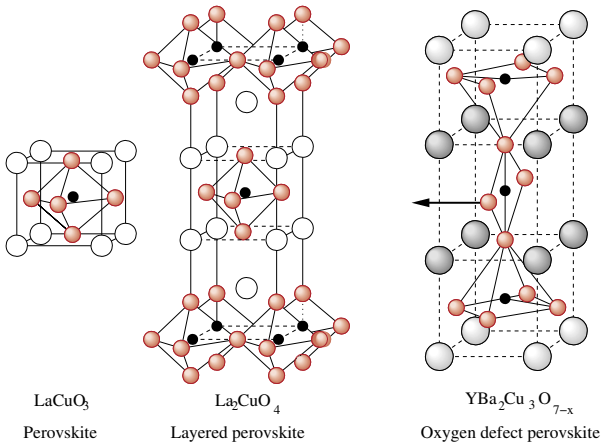

Fig.18: Structure of the Cuprates. The Perovskite structure, at left, is the basic building brick. The middle structure contains the perovskite unit surrounded by CuO_2 planes along the c axis whereas the last one, with oxygen defects, misses oxygen atom composition by a fraction x, as indicated by the arrow. Cu is represented by black spheres, La by white spheres, Oxygen by red spheres, Yttrium by light grey spheres and Barium by dark grey spheres. Adapted from Serway (20)

B. Strong interactions and the Hubbard model

The simplest model concerning the physics of CuO_2 planes is the Hubbard model. It takes into account oxygen role through the presence of a hopping energy (overlap) t of an electron from a Copper atom to its nearest neighbor. It considers also the strong intra-site Coulomb repulsion U between two electrons at the same site.
This model depends on two parameters: the ratio t/U and the number of electrons filling the CuO_2 plane. Despite its simplicity, this model is rather difficult to solve. Nevertheless, the use of optical lattices (32) are providing helpful hints toward finding solutions for the Hubbard model.

Using QG (Quantum Gas) microscopes (32) in ultracold atom experiments has allowed to observe directly and manipulate atoms in lattice sites. Optical lattices are sharpening our capability to simulate strongly correlated systems besides the Hubbard model such as Mott insulator and High-Tc superconductivity.

In the $U \gg t$ case with one electron per site, (half-full band), this model can be cast into a spin Heisenberg Hamiltonian: $H = -J \sum_{<i,j>} \boldsymbol{S}_i \cdot \boldsymbol{S}_j$ with $J = 4t^2/U$.

The interaction between spins $\boldsymbol{S}_i, \boldsymbol{S}_j$ is positive, meaning that neighboring spins at i, j sites tend to align antiferromagnetically. There is no long-range magnetic order at finite temperature in 2D. However, there exist very weak interplane couplings which lead to an antiferromagnetic order at finite temperature (around 300 K for La_2CuO_4)

For a low number of holes (low x doping by Sr in La_2CuO_4) these compounds are insulating and antiferromagnetic at low temperature. Above some doping threshold, superconductivity is observed. Critical temperature varies with doping. It reaches a maximum value for some optimal doping. The compound is said to be "underdoped" or "overdoped" when it is below or above this optimal value.

Concerning the physical origin of the effective attraction between electrons there are two views for these compounds: A "conventional" view claiming that it is still a matter of electron-phonon coupling stronger than in ordinary metals. This perspective is certainly still valid for MgB_2, but difficult to justify in cuprates.
The majority tendency rallies around the idea that magnetism and superconductivity are intimately linked in these compounds. In an ordered magnetic compound, there exist quantum fluctuations around some equilibrium value of the spin order, called "spin waves" that are quantized as "bosons". Similarly to lattice vibrations quantization into "phonons", spin waves can be quantized into "magnons". Phonons and Magnons are elementary excitations with bosonic character.

It is possible to invoke an electron-magnon interaction that would play the same role as the electron-phonon

interaction in the conventional superconductivity functional to produce an effective attraction between electrons.

The isotope effect is an indicator telling whether the effective interaction between electrons is mediated or not by phonons. It appears that for optimally doped compounds, the isotope effect is much lower than for a conventional superconductivity. In fact, for certain compounds, regardless of doping, an isotope effect has been observed.

In BCS theory, Cooper pairs are in an orbital s-state and the half-gap Δ is assumed to be independent of wavevector $\boldsymbol{k}$ (dispersion-less). Moving beyond BCS s-wave pairing and assuming a $\boldsymbol{k}$ function $\Delta(\boldsymbol{k})$ with orbital d-state symmetry since many experimental results agree for a d-type symmetry (28) for the order parameter. Thus we write in the case of d-wave pairing:

$$\Delta(\boldsymbol{k}) = \Delta_0(\cos k_x a - \cos k_y a) \tag{67}$$

such that when $|\boldsymbol{k}| \to 0$ we get $\Delta(\boldsymbol{k}) \propto (k_x^2 - k_y^2)$ like a d-orbital $d_{x^2-y^2}$ function.
This implies, that the gap is zero on some points of the Fermi surface implying incompatibility with a "conventional" explanation of superconductivity (28).

Coherence length is extremely small, about 2 to 3 nm, in many Cuprates. One of the reasons why these superconductors still do not have high-power industrial applications is the weakness of critical currents (by comparison to those in intermetallic compounds based on e.g. Niobium). This is due to their nature. They are ceramics formed by microscopic grain stacking. Each grain is SC and is coupled to its neighbors by a Josephson effect. The critical current is limited by the maximum current through these Josephson links.

APPENDIX A: Superconductivity Timeline

The timeline is displayed in Fig. 19.

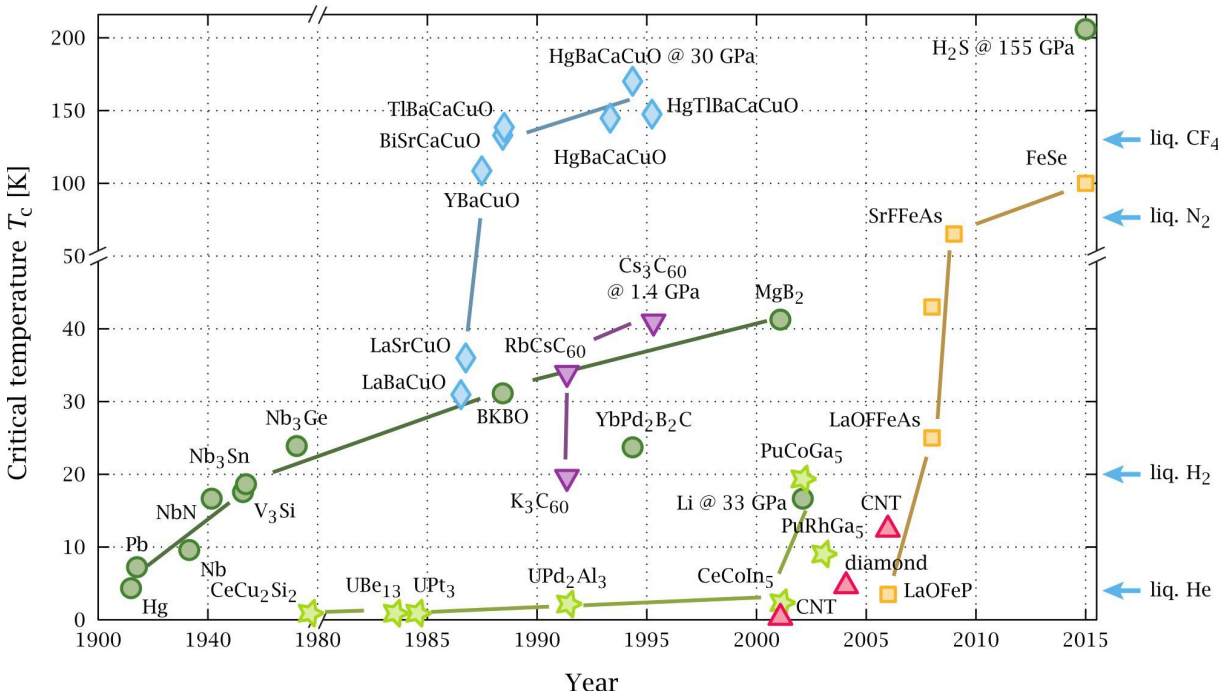

Fig.19: Timeline of superconductivity discoveries with evolution of T_c. Value of pressure in GPa is indicated for materials needing application of pressure to become SC. CNT is Carbon Nano-Tube. Note also the Heavy-Fermion materials such as $CeCu_2Si_2$ and the Fullerene C_{60} compounds. In the case of the pnictide FeSe, T_c reaches 100K and above when the material is a single-layer film grown on $SrTiO_3$ (100) (STO) substrate. Adapted with licence CC BY 4.0 from creativecommons.org

- 1908: Liquefaction of ^{4}He (4.2 K) obtained by Kamerling Onnes in a laboratory in Leiden (Netherlands). During the years that followed, K. Onnes seeks to characterize the behavior of the resistivity of metals at the lowest temperatures ever reached at that time. At high temperature, the resistivity of metals is dominated by the electron-phonon interactions . If T is significantly higher than Debye temperature θ_D, the number of phonons is according to Bose statistics $n(q) = 1/(\exp(\hbar\omega(q)/k_BT) - 1) \approx k_BT/h\omega(q)$ and the resistivity is linear in T. On the other hand, at temperatures much lower than θ_D, the resistivity is dominated by collisions with impurities and lattice defects and tends towards a residual constant proportional to the number of defects. K. Onnes sought to test increasingly pure metals and ended up choosing mercury, which at that time could be almost entirely purified from copper, gold or platinum...

- 1911: Kamerling Onnes discovers superconductivity by proposing to G. Holst to measure the resistivity of mercury in liquid ^{4}He. [Nobel prize] In order to find out whether resistivity is simply small and finite, Kamerlingh-Onnes noticed that during the few hours before ^{4}He total evaporation, no current damping occurred.
  A more precise experiment based on field decay in a solenoid and measured by Nuclear Magnetic Resonance carried out in 1963 by J. File and R. G. Mills (8) indicates at least one hundred thousand years for the current decay in a superconductor. The device was maintained for a year and a half in a SC state.

- 1923: The liquefaction of ^{4}He is obtained in two other laboratories: Berlin and Toronto.

- 1933: Meissner and Ochsenfeld demonstrate the "diamagnetism" of SCs (magnetic flux expulsion).

- 1954: First SC electromagnet (Nb): 0.7 Tesla at 4.2 K

- 1957: Microscopic theory of the SC of metals by Bardeen, Cooper and Schrieffer [Nobel Prize]

- 1962: B. Josephson predicts quantum effects which are used in ultra-sensitive detection of magnetic fields (Josephson Junction, SQUID ...) [Nobel Prize]

- 1970: Launch of a magnetic levitation (Maglev) train project in Japan. The first prototype reached a 400 km/h speed in 1987.

- 1982: First MRI images. The magnetic fields necessary for imaging medical by nuclear magnetic resonance are produced by SC magnets.

- 1986: Bednorz and Müller discover SC in ceramics based on cuprates: $La_{2-x}Ba_xCuO_4$. [Nobel prize]

- 1987: First SC (90 K) above liquid nitrogen temperature (77 K): Y $Ba_2Cu_3O_{7-x}$ called Oxygen deficient perovskite.

- 1995: Reproducible record at 164 K for Hg-Ba-Ca-Cu-O

- 2003: No new reproducible record for 10 years. Several thousands of publications on new SCs with no definitive theoretical interpretation.

- 2008: Discovery of the Iron pnictides such as FeAs (compounds of the nitrogen group) with a T_c of 26 K. The critical temperature is low but the main point in this discovery is that an Iron compound known for its magnetism disagrees with SC.

- 2015: Discovery of Room temperature SC in carbonaceous sulfur hydride H_2S at 287.7 K Kelvin (15C) after its powder was crushed under 155 GPa pressure.

- 2020: A. P. Drozdov *et al.* (33) discover SC near room temperature at 250 K in lanthanum hydride LaH_{10} under high pressure between 150 and 170 GPa.

- 2020: SC is discovered in a thin nickel oxide (nickelate) film.

References

[1] C. Kittel, *Introduction to Solid State Physics*, 8th edition, Wiley, New-York (2005).
[2] L. D. Landau and E. M. Lifshitz, *Statistical Physics, Part1*, Third Edition, Vol. 5, Pergamon, Oxford (1976). E. M. Lifshitz and L.P. Pitaevskii *Statistical Physics, Part2*, Second Edition, Vol. 9, Pergamon, Oxford (1980). L. D. Landau and E. M. Lifshitz, *Quantum Mechanics*, Vol. 3, Pergamon, Oxford (1965).

[3] Hoddeson, Lillian and Vicki Daitch, *True genius: the life and science of John Bardeen*, Joseph Henry Press (Washington, D.C.) (2002), J.R. Schrieffer, *Superconductivity* (W.A. Benjamin, New York, 1964)
[4] H. Ibach and H. Lüth, *Solid-State Physics*, 4th ed., Springer-Verlag, Berlin Heidelberg (2009)
[5] C. J. Pethick and H. Smith, *Bose-Einstein condensation in dilute gases*, 2nd edition, Cambridge University Press (2008).
[6] M. Tinkham, *Introduction to Superconductivity* (Dover Publications, New York, 2nd ed., 2004).
[7] R. Doll and M. Näbauer, Phys. Rev. Lett. **7**, 51 (1961)
[8] J. File and R. G. Mills, Phys. Rev. Lett. **10**, 93 (1963)
[9] G. Parisi, *Mean field theory for spin glasses* in *Disordered Systems and Localization*, Lecture Notes in Physics, **149**, 107-117, Springer-Verlag (1981).
[10] L. P. Gorkov, Sov. Phys. JETP **36**, 1364 (1959)
[11] F. W. J. Olver, D. W. Lozier, R. F. Boisvert, C. W. Clark, *NIST Handbook of Mathematical Functions* Cambridge University Press, New York, (2010)
[12] I. S. Gradstein and I. M. Ryzhik, *Table of Integrals, Series and Products*, Academic Press, New-York (1980).
[13] P. G. De Gennes, *Superconductivity of Metals and Alloys*, Addison-Wesley, Reading (1992)
[14] A. L. Fetter and J. D. Walecka, *Quantum Theory of Many-Particle Systems*, McGraw-Hill, New York (1971).
[15] D. Zubarev, V. Morozov and G. Röpke, *Statistical Mechanics of Non-equilibrium Processes*, Volume 1, Akademie Verlag, Berlin (1996).
[16] D. A. McQuarrie, *Statistical Mechanics*, Harper and Row, New York (1973).
[17] H.B. Callen, *Thermodynamics and an introduction to thermostatistics*, 2nd edition, Wiley, New-York (1985).
[18] F. Reif, *Statistical and Thermal Physics*, McGraw-Hill, New-York (1985).
[19] J. Bardeen, L. N. Cooper, and J. R. Schrieffer, Phys. Rev. **106**, 162 (1957); **108**, 1175 (1957).
[20] R. A. Serway, C. J. Moses and C. A. Moyer, *Superconductivity*, Chapter 12 in *Modern Physics*, Second Edition, W. B. Saunders College Publishing, Fort Worth (1997).
[21] J. Bardeen, Phys. Rev. Lett. **6**, 57 (1961)
[22] J. R. Schrieffer and J. W. Wilkins, Phys. Rev. Lett. **10**, 17, (1963). See also: E. Burstein and S. Lundqvist, *Tunneling Phenomena in Solids*, Chapters 21 and 24, Plenum Press and Springer, New-York (1969).
[23] C. B. Duke, *Tunneling in Solids*, in *Solid State Physics*, Supplement 10, Academic, New-York (1969).
[24] S.M. Sze, *Physics of Semiconductor Devices*, 2nd edition, Wiley, New York (1981)
[25] R.P. Feynman, R. Leighton and M. Sands, *Feynman Lectures on Physics*, Vol 3: *Quantum Mechanics*, Chapter 21, Addison-Wesley (1965).
[26] C. C. Grimes and S. Shapiro, Phys. Rev. **169**, 397 (1968)
[27] K. Biedermann, A. Chrestin, T. Matsuyama, and U. Merkt, Phys. Rev. **B 63**, 144512 (2001)
[28] D. Koelle, R. Kleiner, F. Ludwig E. Dantsker and J. Clarke, *High-transition-temperature superconducting quantum interference devices* Rev. Mod. Physics, **71**, No. 3, p 631 (1999).
[29] R. L. Fagaly, Rev. Sci. Inst. **77**, 101101 (2006)
[30] P. A Lee, N. Nagaosa, and X-G Wen, *Doping a Mott insulator: Physics of high-temperature superconductivity*, Rev. Mod. Phys. **78**, 17. (2006)
[31] C. M. Varma, *Colloquium: Linear in temperature resistivity and associated mysteries including high temperature superconductivity*, Rev. Mod. Phys. **92**, 031001 (2020)
[32] I. Bloch, J. Dalibard, and W. Zwerger, *Many-body physics with ultracold gases*, Rev. Mod. Phys., **80**, 885-964, (2008).
[33] A. P. Drozdov, M. I. Eremets, I. A. Troyan, V. Ksenofontov, and S. I. Shylin, Nature (London) **525**, 73 (2015)